

PV–PWR–4 power board

Rev A1 | 2026–10–04 | Schematics authored by Chinmoy Bhuyan | PV–P100/110: 4 interleaved FSBB phases, gate drive, sensing, ports, aux (D–044)

01 PC connector, command buffers, status

Interfaces PC (2x32) to the control board, 3.3 -> 5 V command buffers, RDY / FLT_N, +5 V supervisor, port status glue, BUS– star points

File: sheets/01_pc.kicad_sch

02 Phase 1 power stage, bias, sensing

Legs A/B (2 x SG2M040170HJ per switch), decoupling, dampers, bank share, inductor + STK–HQ/A 75, NTC inputs, gate bias (gdrv.bias_phase)

File: sheets/02_ph1_power.kicad_sch

03 Phase 1 leg A gate drive (S1, S2)

gdrv.channel() rev 7, bias='ext': NSI6651ASC–Q1 +18/–3.5 V, 3.75/2.5 ohm per device, Miller clamps, SC booster 22R, interlock + dead–time stretch, DESAT to the drain

File: sheets/03_ph1_gda.kicad_sch

04 Phase 1 leg B gate drive (S3, S4)

gdrv.channel() rev 7, bias='ext': NSI6651ASC–Q1 +18/–3.5 V, 3.75/2.5 ohm per device, Miller clamps, SC booster 22R, interlock + dead–time stretch, DESAT to the drain

File: sheets/04_ph1_gdb.kicad_sch

05 Phase 2 power stage, bias, sensing

Legs A/B (2 x SG2M040170HJ per switch), decoupling, dampers, bank share, inductor + STK–HQ/A 75, NTC inputs, gate bias (gdrv.bias_phase)

File: sheets/05_ph2_power.kicad_sch

06 Phase 2 leg A gate drive (S1, S2)

gdrv.channel() rev 7, bias='ext': NSI6651ASC–Q1 +18/–3.5 V, 3.75/2.5 ohm per device, Miller clamps, SC booster 22R, interlock + dead–time stretch, DESAT to the drain

File: sheets/06_ph2_gda.kicad_sch

07 Phase 2 leg B gate drive (S3, S4)

gdrv.channel() rev 7, bias='ext': NSI6651ASC–Q1 +18/–3.5 V, 3.75/2.5 ohm per device, Miller clamps, SC booster 22R, interlock + dead–time stretch, DESAT to the drain

File: sheets/07_ph2_gdb.kicad_sch

08 Phase 3 power stage, bias, sensing

Legs A/B (2 x SG2M040170HJ per switch), decoupling, dampers, bank share, inductor + STK–HQ/A 75, NTC inputs, gate bias (gdrv.bias_phase)

File: sheets/08_ph3_power.kicad_sch

09 Phase 3 leg A gate drive (S1, S2)

gdrv.channel() rev 7, bias='ext': NSI6651ASC–Q1 +18/–3.5 V, 3.75/2.5 ohm per device, Miller clamps, SC booster 22R, interlock + dead–time stretch, DESAT to the drain

File: sheets/09_ph3_gda.kicad_sch

10 Phase 3 leg B gate drive (S3, S4)

gdrv.channel() rev 7, bias='ext': NSI6651ASC–Q1 +18/–3.5 V, 3.75/2.5 ohm per device, Miller clamps, SC booster 22R, interlock + dead–time stretch, DESAT to the drain

File: sheets/10_ph3_gdb.kicad_sch

11 Phase 4 power stage, bias, sensing

Legs A/B (2 x SG2M040170HJ per switch), decoupling, dampers, bank share, inductor + STK–HQ/A 75, NTC inputs, gate bias (gdrv.bias_phase)

File: sheets/11_ph4_power.kicad_sch

12 Phase 4 leg A gate drive (S1, S2)

gdrv.channel() rev 7, bias='ext': NSI6651ASC–Q1 +18/–3.5 V, 3.75/2.5 ohm per device, Miller clamps, SC booster 22R, interlock + dead–time stretch, DESAT to the drain

File: sheets/12_ph4_gda.kicad_sch

13 Phase 4 leg B gate drive (S3, S4)

gdrv.channel() rev 7, bias='ext': NSI6651ASC–Q1 +18/–3.5 V, 3.75/2.5 ohm per device, Miller clamps, SC booster 22R, interlock + dead–time stretch, DESAT to the drain

File: sheets/13_ph4_gdb.kicad_sch

14 Port A power path (pv)

Varistor Y + monitor

File: sheets/14_leanA_power.kicad_sch

15 Port A sensing and interlocks

References, BUS– shunt amplifier + hold–off / OC windows, dividers, dV amplifier, polarity and dV interlocks

File: sheets/15_leanA_ctrl.kicad_sch

16 Port A coil drives

Contactors: interlock logic, fail–safe high side, economiser; passive bank bleeder

File: sheets/16_leanA_drive.kicad_sch

16 Port B power path (battery)

CAUTION: not for use as a design_check() asserts every number on each build (outputs: PV–PWR–4 design_check.txt), drafted @ few days from design, do not use at build time. Domains: LIVE (BUS– = GND = AGND), GD_PxSy gate islands (functional: NSI6651, T_BIAS4), PE (AIN pads, inductor cores, heatsink NTC lugs, Y / MOV3 / MD crossings), SELV (AUX–T1 reinforced winding -> J_SELV). Coat to PD1. Shared heatsink: Tj 110 C at 45 C inlet (R_sa target); heatsink OT trip must be <= 90 C. Dead time 211–554 ns. Sensor delay 0.33 us, error 1.8 % at 45 A after calibration. Live 24 V 23.7 W (economiser) <= aux live rating 24.0 W; module aux 80 W. Full fan speed is only guaranteed while the converter is switching.

File: sheets/16_leanB_power.kicad_sch

17 Port B sensing and interlocks

References: PV–PWR–4 design_check.txt, drafted @ few days from design, do not use at build time. polarity and dV interlocks

File: sheets/17_leanB_ctrl.kicad_sch

18 Port B coil drives

Contactors + precharge relay: interlock logic, fail–safe high side, economiser; passive bank bleeder

File: sheets/18_leanB_drive.kicad_sch

18 Terminals, EMI filter, insulation test

DC terminals, nanocrystalline CM ring per port, X / Y capacitors, board entries (press–fit), PE links, heatsink, insulation monitor (port.lean_imd)

File: sheets/18_terminals_imd.kicad_sch

19 Live 5 V / 3.3 V, SELV connector

+24V (aux live) -> +5V and +5V_GD (LMR38020), +3V3 (TPS62130 from +5V_GD ramp–up and discharge), +5V_GD supervisor, SELV connector to the control board

File: sheets/19_live5v.kicad_sch

20 AUX75 input taps, OR, film

One tap per port (+ pole, terminal side), return = BUS–: 2 x 20R R pulse–rated ahead of the 1 A / 1000 VDC fuse, BYG10Y OR; 2 x 3.3 uf / 1300 V film = surge absorber (AUX–HV rev C1 input block)

File: sheets/20_aux75_input.kicad_sch

21 AUX75 start–up, brown–in, OV, timer

As AUX–HV rev C1 sheet 02: 3 x BSS126 cascode, TPS3710 brown–in 206–215 V, TLV3202 OV lockout 1107–1162 V and overload timer 35–80 ms

File: sheets/21_aux75_start.kicad_sch

22 AUX75 PWM, SIC, clamp, AUX–T1

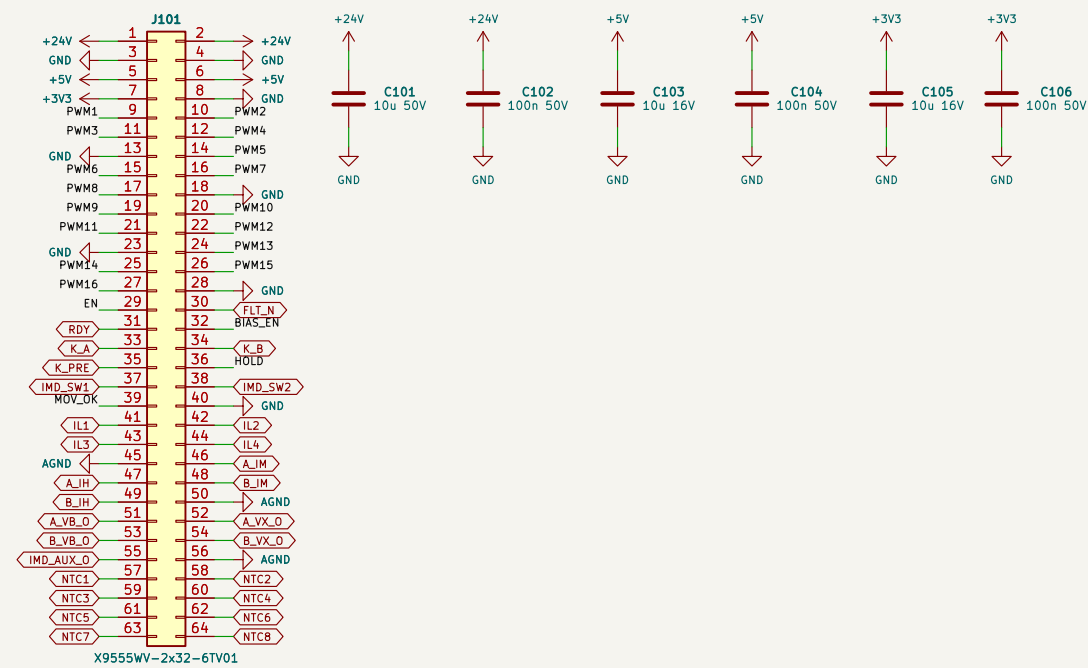
UCC28C59–Q1 64.9 kHz, IV2Q171R0D7Z, Lp 0.485 mH, Np:Nlive:Nselv 48:8:9; live 24 V regulated by the FB divider (no opto); VDD from the live 24 V through the BC817 follower; foldback detector on the live winding

File: sheets/22_aux75_pwm.kicad_sch

Not bench–validated. Values read from sim/out/*.json; checks: design_check()
Sheet 1
Schematics authored by Chinmoy Bhuyan
Live board referenced to BUS–; gate islands functional; SELV winding reinforced; HV–PE basic

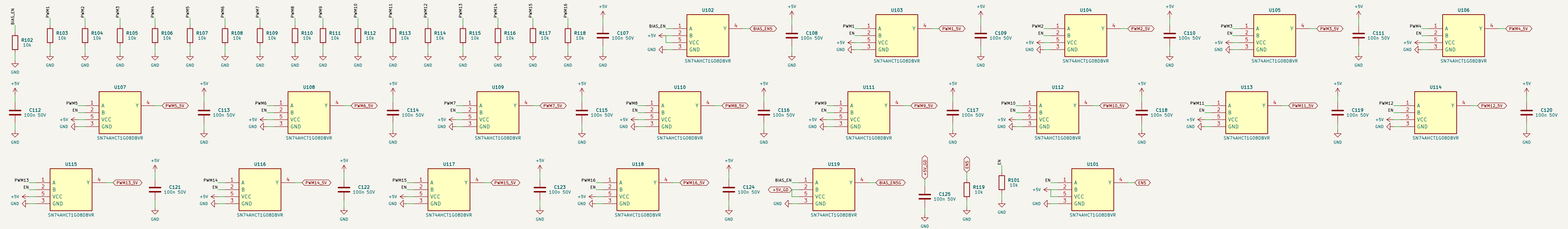
Sheets: 1
File: F
Title
Size: 6
KiCad

PC CONNECTOR (INTERFACES.PC), LIVE



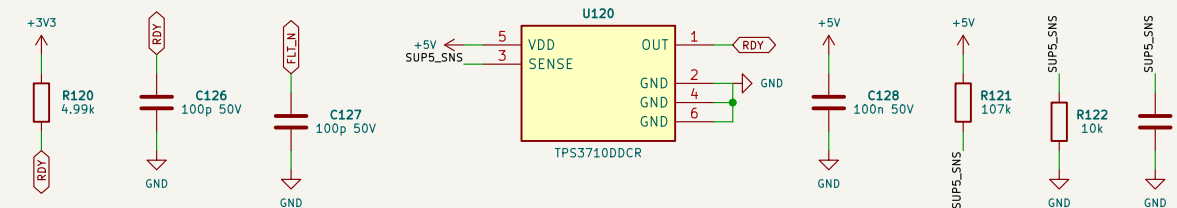
Both boards live: GND = AGND = BUS-, Supplies +24V / +5V / +3V3 go TO the control board. Commands are 3.3 V logic after the control board's trip latch; every command input is pulled low here (open = off).

COMMAND INPUTS: PULL-DOWNS AND 3.3 -> 5 V BUFFERS (GDRV.INPUT_BUFFERS)



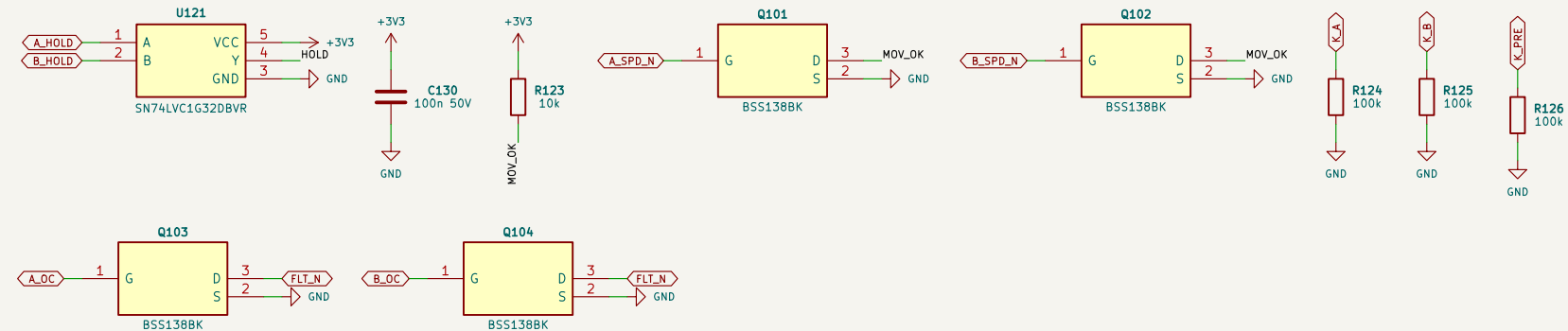
NS16651 inputs are specified at VCC1 = 5 V only (VINH <= 3.5 V, p7) and the SN6505B EN needs 0.7 VCC: one SN74AHC1G08 (VIH 2.0 V) per line. Each PWM is ANDed with EN here as well, so a low EN removes every command within 8 ns, ahead of the drivers' RST/EN filter (0.48–0.8 us).

RDY (WIRED-AND) AND FLT_N (WIRED-OR)



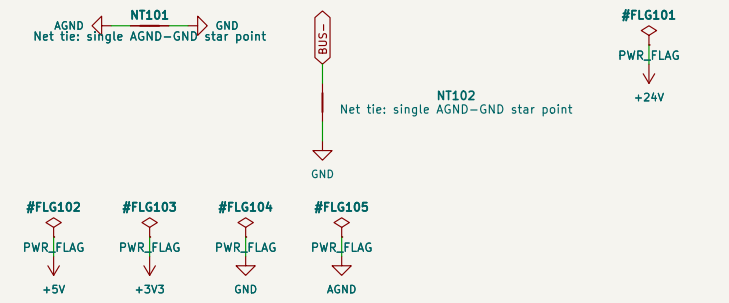
RDY: every NSI6651 RDY (VCC1 and VCC2 UVLO), the +5V and +5V_GD supervisors and the aux status (live 24 V UV), pulled up HERE to +3V3 (4.99k). FLT_N: every NSI6651 /FLT + port 0C; pull-up on the control board. +5V supervisor TPS3710 107k/10k; RDY low below 4.52-4.69 V (AHCT VCC >= 4.5 V, bias headroom).

PORT STATUS TO PC: HOLD, MOV_OK, PORT OVER-CURRENT INTO FLT_N



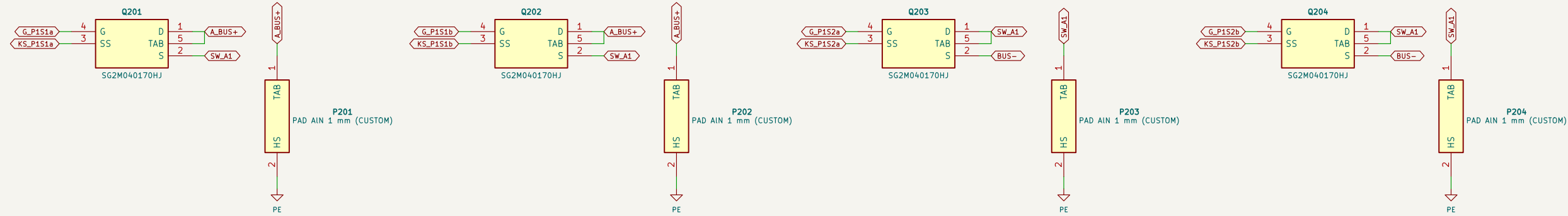
HOLD = A_HOLD OR B_HOLD (|I_port| >= 0.97-1.03 kA, contactor held). MOV_OK high only while BOTH varistor monitor loops conduct (each reads 'open' below ~200 V on its port). Port OC (|I| >= 387-413 A, lean_shunt) pulls FLT_N low (open drain): the control board's latch trips without firmware (aR=fuse gap, port_spec). K_A / K_B / K_PRE pulled low: open = coil off (the hardware interlocks follow on the port sheets).

GROUND STAR POINTS



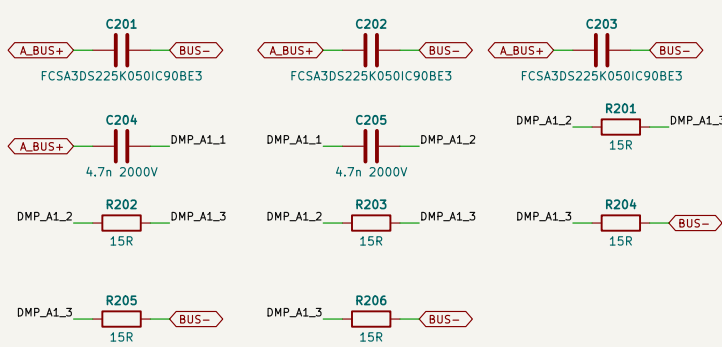
AGND meets GND at one point: GND meets BUS— at one point next to the port shunts (controller ground = BUS—). Layout: gate-drive primaries decoupled to GND locally.

PHASE 1 LEG A: S1 (TOP) / S2 (BOTTOM), 2 X SG2M040170HJ EACH



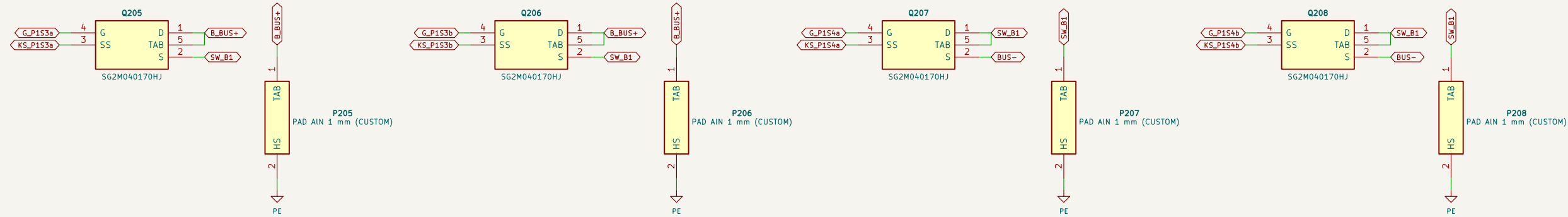
Kelvin source (pin 3) of each device to its own driver COM star (gate-drive sheet). Each tab on its own. AlN ceramic (aluminum nitride) pad to the EARTHED heatsink (basic HV-PE, B2). LAYOUT (IC-22): milled slot between drain pin 1 and pins 2-4; equal drain / source busbar paths for the two devices of a switch.

PHASE 1 LEG A DECOUPLING + RC DAMPER



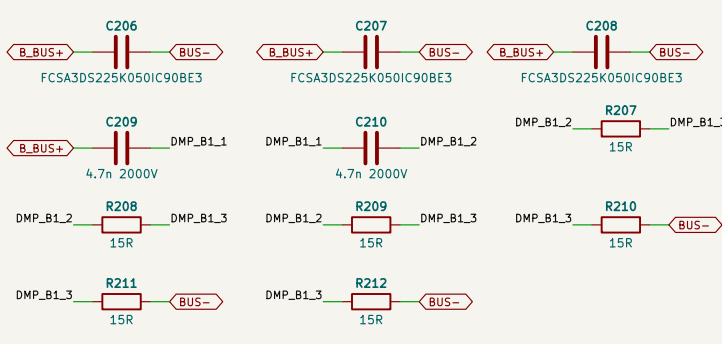
3 x FCSA3DS225 at the S1/S2 pins: commutation loop ≤ 20 nH, bank ≤ 15 nH from the leg (LAYOUT, leg deck).
Damper (PVR-02): 2 x (3 x 15R CRCW2512-HP) + 2 x 4.7n 2 kV COG in series, loop at the pins ≤ 3 nH.

PHASE 1 LEG B: S3 (TOP) / S4 (BOTTOM), 2 X SG2M040170HJ EACH



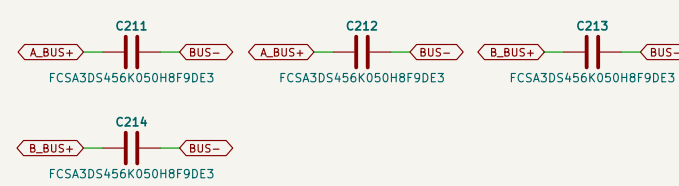
Kelvin source (pin 3) of each device to its own driver COM star (gate-drive sheet). Each tab on its own AlN ceramic (aluminum nitride) pad to the EARTHED heatsink (basic HV-PE, B2). LAYOUT (IC-22): milled slot between drain pin 1 and pins 2-4; equal drain / source busbar paths for the two devices of a switch.

PHASE 1 LEG B DECOUPLING + RC DAMPER



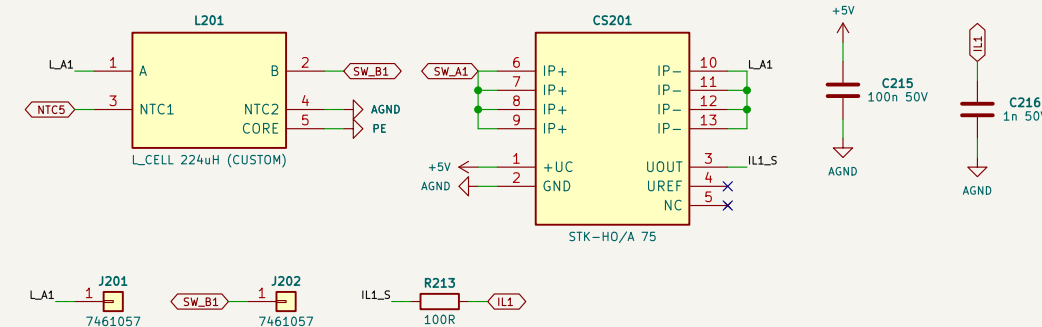
3 x FCSA3DS225 at the S3/S4 pins: commutation loop ≤ 20 nH, bank ≤ 15 nH from the leg (LAYOUT, leg deck).
Damper (PVR-02): 2 x (3 x 15R CRCW2512-HP) + 2 x 4.7n 2 kV COG in series, loop at the pins ≤ 3 nH.

PHASE 1 SHARE OF THE PORT BANKS



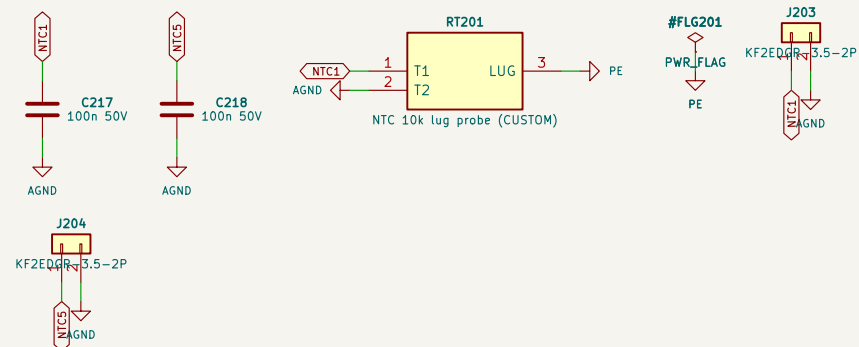
2 x FCSA3DS456 (45 uF) on A_BUS+ and 2 on B_BUS+, at this phase's legs: the banks of all phases form the port A / port B DC links. Bleeders: one string per port on the port sheets.

PHASE 1 INDUCTOR (CHASSIS, CUSTOM) AND CURRENT SENSOR



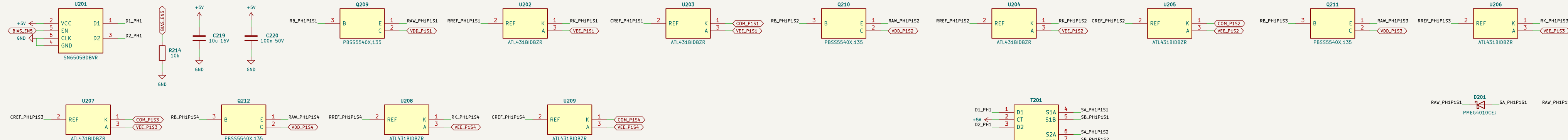
Lead A from SW_A1 passes the STK-HO/A 75 U-primary (pins 6-9 in, 10-13 out: + = SW_A -> SW_B) to stud L_A1; lead B on stud SW_B1. IL1 = 2.5 V + 10.667 mV/A (Vref 2.48-2.52 V), 0.5-4.5 V over +/-187.5 A; 100R + 1 nF at the source. Primary at the switch node: functional barrier (4 kV rms, 8 kV impulse).

PHASE 1 NTC INPUTS (NTC TO AGND, BIASED ON THE CONTROL BOARD)

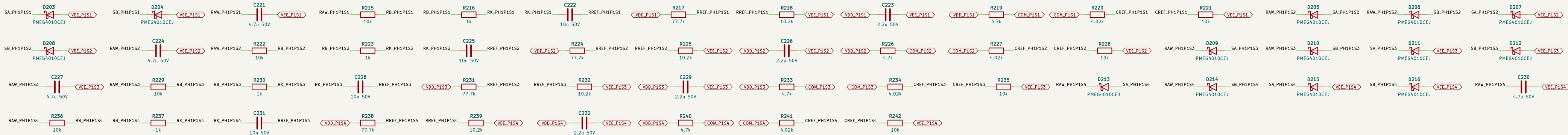


NTC1 = heatsink section of phase 1: lug probe on the EARTHED heatsink. >= 2.5 kV rms lead-to-lug (R-10).
NTC5 = inductor winding NTC (functional to the winding). 100 nF at the header against switch-node noise.

PH1: GATE BIAS (PER PHASE)



Gate-bias transformer per phase (sim/out/magnetics/design_gdrv_blas_transformer.json): DMEGC EP17 set (DMR44), ungapped, 4+4:24 x4, ratio 1:6, barrier 1.77 pF high-side winding to primary / 3.43 pF adjacent sections, functional insulation PD-free >= 1.75 kVpc; 2.18 USD (1 k, ESTIMATE)



SN6505B push-pull 420 kHz from +5V, BIAS_EN high = on; T_BIAS4 custom, functional insulation (PD-free ≥ 1.75 kVpk, ≤ 5 pF per secondary). Per channel: bridge, PNP regulator VDD-VEE, ATL431 COM split

Not bench-validated. Values read from sim/out/*.json; checks: design_check()
Sheet 3

Schematics authored by Chinmoy Bhuyan
Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basis

Sheet: /02 Phase 1 power stage, bias, sensing/

File: 02_ph1_power.kicad_sch
Title: PV-PWR-4 power board - Phase 1 power stage, bias, sensing

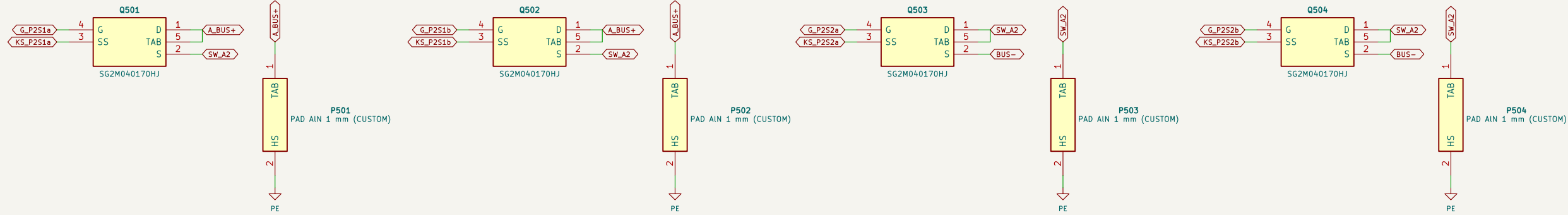
Size: A2	Date: 2026-10-04	Rev: A1
KiCad E.D.A. 10.0.6		Id: 3/26

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Figure 1: Schematic representation of the genetic constructs. The figure shows two main constructs, Q414 and Q415, and their various derivatives. Q414 contains BTB_P154, Q414, and MMBT3904_215. Q415 contains BHC_P154, Q415, and MMBT3906_215. Derivatives include DX_P154 (D430, BZXB4-810), BTZ_P154 (R462, 224), BTB_P154 (R463, 100k), BTC_P154 (R464, 2.2k), BHB_P154 (R465, 10k), BHC_P154 (R466, 220R), BG_P154 (D431, BZXB4-810), and BG_P154 (R467, 1k). Each construct is represented by a circle with a central node and lines connecting to various elements. The elements are color-coded: blue for BTB_P154, green for BHC_P154, red for R462, R463, R465, R466, R467, and yellow for D430, D431, BZXB4-810, and VEE_P154. The elements are labeled with their respective names and sizes.

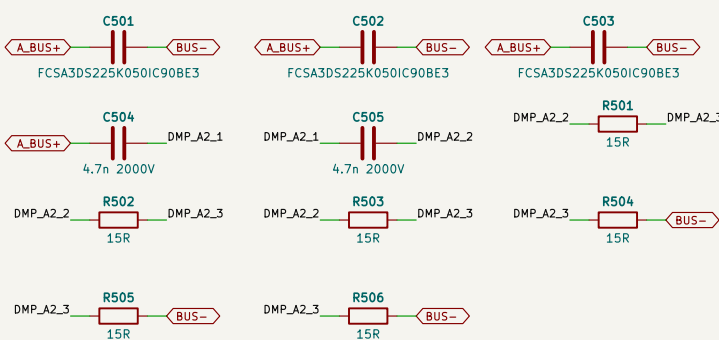
Gate and Kelvin of each device as coupled pairs of equal length (+/-10 %) from the COM star: Ron, Roff, Kelvin R, 10k, Zeners at each device. Clamp FET + C_{VL} at its device's gate-Kelvin pins, loop <= 1 nH. DESAT string >= 2.5 mm pad gap per diode; board coated to PD1 (IC-22). Miller hold (ngspice, gdrv): die +1.41 V vs VGS(th) min 1.94 V at 175 C - BENCH TEST. Driver input side and bias primary decoupled to GND locally (CM current).

PHASE 2 LEG A: S1 (TOP) / S2 (BOTTOM), 2 X SG2M040170HJ EACH



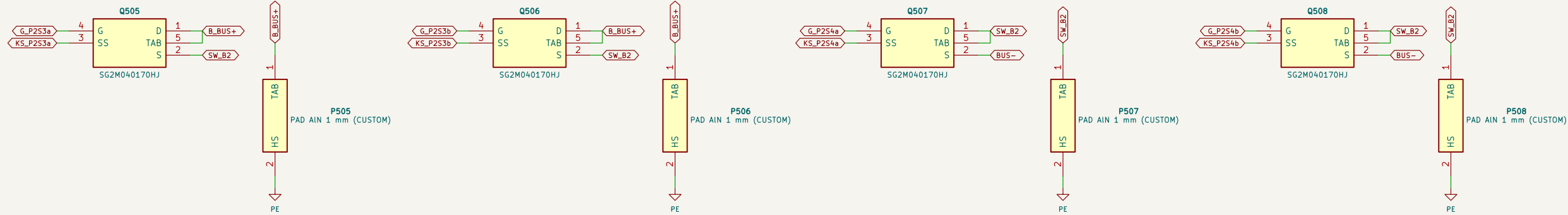
Kelvin source (pin 3) of each device to its own driver COM star (gate-drive sheet). Each tab on its own
AlN ceramic (aluminium nitride) pad to the EARTHED heatsink (basic HV-PE, B2). LAYOUT (IC-22): milled slot between drain pin 1 and
pins 2-4; equal drain / source busbar paths for the two devices of a switch.

PHASE 2 LEG A DECOUPLING + RC DAMPER



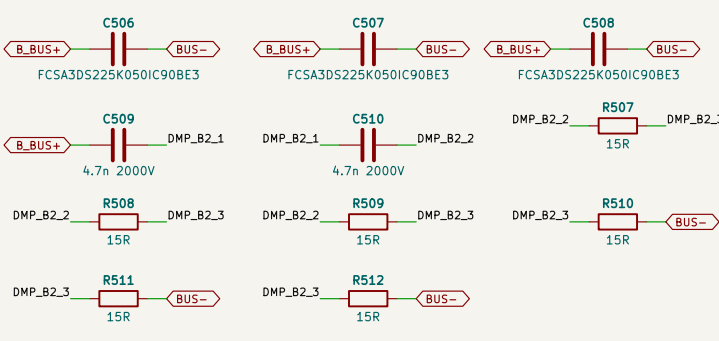
3 x FCSA3DS225 at the S1/S2 pins: commutation loop <= 20 nH, bank <= 15 nH from the leg (LAYOUT, leg deck).
Damper (PVR-02): 2 x (3 x 15R CRCW2512-HP) + 2 x 4.7n 2 kV CCG in series, loop at the pins <= 3 nH.

PHASE 2 LEG B: S3 (TOP) / S4 (BOTTOM), 2 X SG2M040170HJ EACH



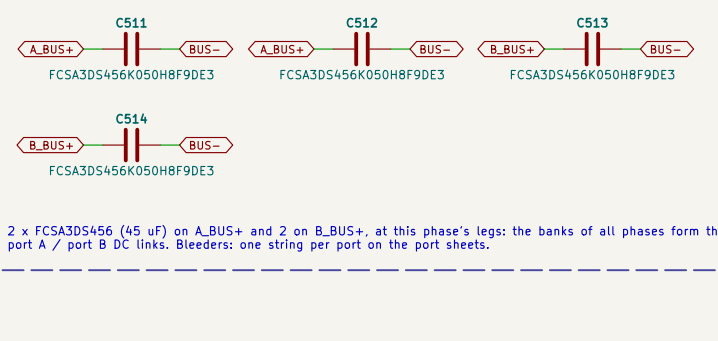
Kelvin source (pin 3) of each device to its own driver COM star (gate-drive sheet). Each tab on its own
AlN ceramic (aluminium nitride) pad to the EARTHED heatsink (basic HV-PE, B2). LAYOUT (IC-22): milled slot between drain pin 1 and
pins 2-4; equal drain / source busbar paths for the two devices of a switch.

PHASE 2 LEG B DECOUPLING + RC DAMPER



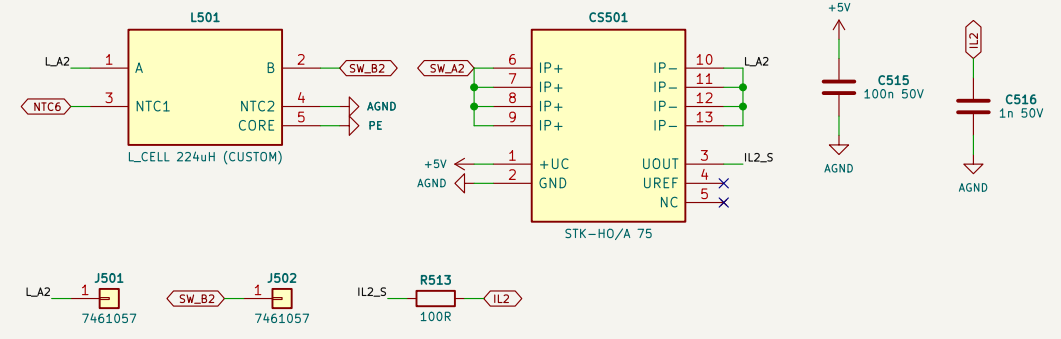
3 x FCSA3DS225 at the S3/S4 pins: commutation loop <= 20 nH, bank <= 15 nH from the leg (LAYOUT, leg deck).
Damper (PVR-02): 2 x (3 x 15R CRCW2512-HP) + 2 x 4.7n 2 kV CCG in series, loop at the pins <= 3 nH.

PHASE 2 SHARE OF THE PORT BANKS



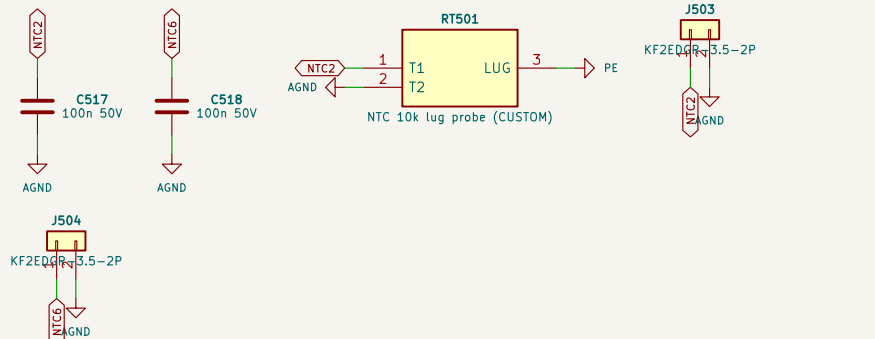
2 x FCSA3DS456 (45 uF) on A.BUS+ and 2 on B.BUS+, at this phase's legs: the banks of all phases form the
port A / port B DC links. Bleeders: one string per port on the port sheets.

PHASE 2 INDUCTOR (CHASSIS, CUSTOM) AND CURRENT SENSOR



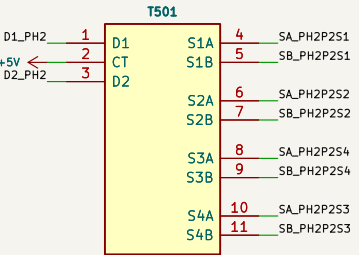
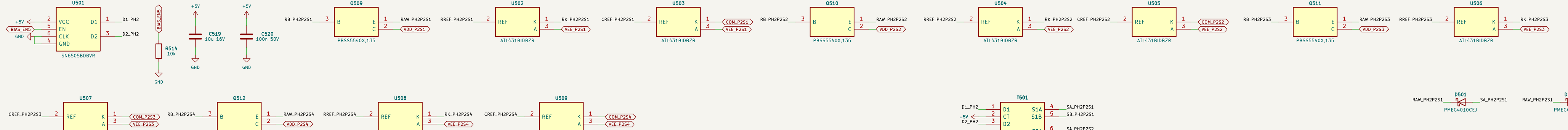
Lead A from SW_A2 passes the STK-HO/A 75 U-primary (pins 6-9 in, 10-13 out: + = SW_A -> SW_B) to stud
LA2; lead B on stud SW_B2; IL2 = 2.5 V + 10.6mV mV/A (Vref 2.48-2.52 V), 0.5-4.5 V over +/-187.5 A;
100R + 1 nF at the source. Primary at the switch node: functional barrier (4 kV rms, 8 kV impulse).

PHASE 2 NTC INPUTS (NTC TO AGND, BIASED ON THE CONTROL BOARD)

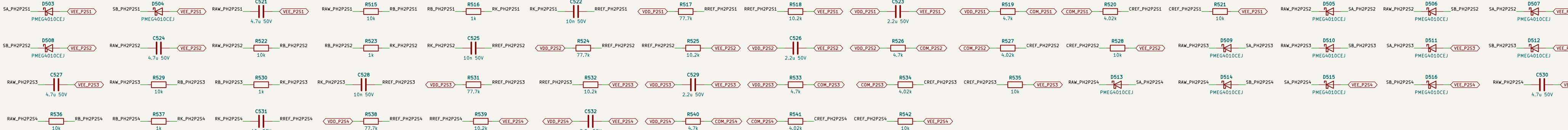


NTC2 = heatsink section of phase 2; lug probe on the EARTHED heatsink, >= 2.5 kV rms lead-to-lug (R-10).
NTC1 = inductor winding NTC (functional to the winding), 100 nF at the header against switch-node noise.

PH2: GATE BIAS (PER PHASE)



Gate-bias transformer per phase (sim/out/magnetics/design_gdrv_bias_transformer.json): DMEGC EP17 set (DMR44), ungapped, 4+4x24 x4, ratio 1:6, barrier 1.77 pF high-side winding to primary / 3/43 pF adjacent sections, functional insulation PD-free >= 1.75 kVpk; 2.18 USD (1 k. ESTIMATE)



SM6505B push-pull 420 kHz from +5V, BIAS_EN high = on; T_BIAS4 custom, functional insulation
(PD-free >= 1.75 kVpk, <= 5 pF per secondary). Per channel: bridge, PNP regulator VDD-VEE, ATL431 COM split.

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P251: DRIVER AND 2 GATES

The schematic diagram illustrates the electrical connections for the P251 driver and two gates. The central component is the U602 MOSFET driver (NSI6651ASC-Q15WR), which is connected to a +5V supply and has pins for INP, IN, EN, FLT, RDY, TEST, GND, VCC1, VCC2, OUTH, OUTL, CLAMP, DESAT, GND2, VEE2, and GND1. The driver is connected to two PMV30E9NR MOSFETs (Q601 and Q602) via a common gate pin (G_P251). The MOSFETs are connected to a +5V supply and have pins for VDD, GND, and VEE. The schematic includes various components like resistors (R605, R606, R607, R608, R609, R610, R611, R612, R613, R614, R615, R616, R617, R618, R619, R620), capacitors (C607, C608, C609, C610, C611, C612, C613), and diodes (D602, D603, D604, D605, D606). The diagram is labeled 'P251: DRIVER AND 2 GATES'.

Per device: Ron/Reff, 0.5R the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.
Miller clamp: N-MOSFET clamp (PMV30E9NR) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P251: MILLER CLAMP CONTROL

The diagram illustrates the Miller Clamp Control circuit for the P251. It features two PMOS transistors, Q604 (MMBT3906) and Q605 (MMBT3904), which are the main switching elements. Their gates are connected to a common clamp node. This clamp node is also connected to the gates of four NMOS transistors: D608, D609, D610, and D611. The clamp node is further connected to a network of resistors (R621, R622, R623, R624, R625) and capacitors (C614, C615). The circuit is powered by VDD_P251 and VEE_P251. The output of the clamp is connected to the gates of NMOS transistors Q604 and Q605. The diagram is labeled 'P251: MILLER CLAMP CONTROL'.

P2S1: SHORT-CIRCUIT BOOSTER

The diagram illustrates the P2S1 Short-Circuit Booster circuit. It consists of two main sections, one for BTB_P2S1 and one for BHB_P2S1. Each section features a MOSFET (Q606 and Q607) with a gate connected to a 10V source (BTB_P2S1_10V and BHB_P2S1_10V) through a 10k resistor (R627, R628). The gates are also connected to a 22k resistor (R629) and a 220R resistor (R631). The drains are connected to a 22k resistor (R627) and a 220R resistor (R631). The sources are connected to a 10k resistor (R630) and a 10k resistor (R628). The gates are also connected to a 10k resistor (R630) and a 10k resistor (R628). The gates are also connected to a 10k resistor (R630) and a 10k resistor (R628).

BTB_P2S1 Section:

- Gate: BTB_P2S1_10V (10V)
- Gate Resistor: R627 (10k)
- Gate Resistor: R628 (22k)
- Gate Resistor: R629 (220R)
- Gate Resistor: R630 (10k)
- Gate Resistor: R631 (10k)
- Gate Resistor: R632 (10k)
- Gate Resistor: R633 (10k)
- Gate Resistor: R634 (10k)
- Gate Resistor: R635 (10k)
- Gate Resistor: R636 (10k)
- Gate Resistor: R637 (10k)
- Gate Resistor: R638 (10k)
- Gate Resistor: R639 (10k)
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- Gate Resistor: R659 (10k)
- Gate Resistor: R660 (10k)
- Gate Resistor: R661 (10k)
- Gate Resistor: R662 (10k)
- Gate Resistor: R663 (10k)
- Gate Resistor: R664 (10k)
- Gate Resistor: R665 (10k)
- Gate Resistor: R666 (10k)
- Gate Resistor: R667 (10k)
- Gate Resistor: R668 (10k)
- Gate Resistor: R669 (10k)
- Gate Resistor: R670 (10k)
- Gate Resistor: R671 (10k)
- Gate Resistor: R672 (10k)
- Gate Resistor: R673 (10k)
- Gate Resistor: R674 (10k)
- Gate Resistor: R675 (10k)
- Gate Resistor: R676 (10k)
- Gate Resistor: R677 (10k)
- Gate Resistor: R678 (10k)
- Gate Resistor: R679 (10k)
- Gate Resistor: R680 (10k)
- Gate Resistor: R681 (10k)
- Gate Resistor: R682 (10k)
- Gate Resistor: R683 (10k)
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- Gate Resistor: R687 (10k)
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- Gate Resistor: R690 (10k)
- Gate Resistor: R691 (10k)
- Gate Resistor: R692 (10k)
- Gate Resistor: R693 (10k)
- Gate Resistor: R694 (10k)
- Gate Resistor: R695 (10k)
- Gate Resistor: R696 (10k)
- Gate Resistor: R697 (10k)
- Gate Resistor: R698 (10k)
- Gate Resistor: R699 (10k)
- Gate Resistor: R700 (10k)

BHB_P2S1 Section:

- Gate: BHB_P2S1_10V (10V)
- Gate Resistor: R627 (10k)
- Gate Resistor: R628 (22k)
- Gate Resistor: R629 (220R)
- Gate Resistor: R630 (10k)
- Gate Resistor: R631 (10k)
- Gate Resistor: R632 (10k)
- Gate Resistor: R633 (10k)
- Gate Resistor: R634 (10k)
- Gate Resistor: R635 (10k)
- Gate Resistor: R636 (10k)
- Gate Resistor: R637 (10k)
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- Gate Resistor: R691 (10k)
- Gate Resistor: R692 (10k)
- Gate Resistor: R693 (10k)
- Gate Resistor: R694 (10k)
- Gate Resistor: R695 (10k)
- Gate Resistor: R696 (10k)
- Gate Resistor: R697 (10k)
- Gate Resistor: R698 (10k)
- Gate Resistor: R699 (10k)
- Gate Resistor: R700 (10k)

P2S2: LOGIC SIDE

The schematic illustrates the logic side of the P2S2 module, centered around the U603 (SN74LVC1G17DBVR) inverter buffer. The input (A) is connected to INP_P2S2 via a 10k resistor (R636). The output (Y) is connected to DTB_P2S2. The circuit also includes pull-up resistors (R637, R639) and pull-down resistors (R638, R640) for various signals. Power supply connections for +5V and GND are shown for multiple pins.

Legend:

- IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p (Interlock + min dead time = max(firmware, hardware)). Bias PG (active low) pulls Rdy low.

P252: DRIVER AND 2 GATES

U604 Pinout:

Pin	Signal	Pin	Signal
1	VEE_P252	9	GND1
2	DK_P252	10	CLAMP
3	COM_P252	11	RST/EN
4	OUTL_P252	12	RDY
5	OUTL_P252	13	FLT
6	OUTL_P252	14	INN_P252
7	CLAMP_P252	15	VCC1
8	VEE_P252	16	INN_P252

NSI6651ASC-Q1SWR

Component Values:

- R640: 0R
- R641: 0R
- R642: 1k
- R643: 3.75R
- R644: 2.5R
- R645: 10k
- R646: 0.5R
- R647: 4.7R
- R648: 22R
- R649: 3.75R
- R650: 2.5R
- R651: 10k
- R652: 0.5R
- R653: 4.7R
- R654: 22R
- R655: 150R

Capacitor Values:

- C621: 10u 50V
- C622: 100n 50V
- C623: 22u 25V
- C624: 22u 25V
- C625: 100n 50V
- C626: 1u 50V
- C627: 1u 50V

Diode Values:

- D620: BZX84-B20.215
- D621: BZX84-B4V7
- D622: PMEG4010CE
- D623: BZX84-B20.215
- D624: BZX84-B4V7
- D625: PMEG4010CE

Other Components:

- Q609: PMV30ENEAR
- Q610: PMV30ENEAR
- Q611: PMV30ENEAR

Per device: Ron/Off, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V on its own Kelvin net. Miller hold: -N-MOSFET clamp (PMV30ENEA) gate-pin to a local VEE (1u on the Kelvin pin). LATOUT: FET + 1u at the device, loop <= 1 nH.

[illegible]

P2S2: SHORT-CIRCUIT BOOSTER

The circuit diagrams illustrate the P2S2 Short-Circuit Booster configuration. The top section shows two MOSFET drivers: one for Q614 (MMBT3904, 215) and another for Q615 (MMBT3906, 215). The bottom section shows four current sources: DX_P2S2 (BZX84-B10, 22k), BTZ_P2S2 (R663, 100k), BTB_P2S2 (R664, 2.2k), and B7B_P2S2 (R665, 10k). The bottom right shows two more current sources: B7C_P2S2 (R666, 220R) and B7B_P2S2 (R667, 1k).

DESAT > 10.3V -10.9 V (above the 9.8 V maximum trip) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V per device AQ3400A + 22R pull the gate to VEE; BG holds -3 us (1k + FET gates).

Figure 1: Pin connections for the Q616 module. The diagram shows the internal connections for various pins. COM_P252 is connected to COM (pin 4) and D632 (pin 1). D632 is connected to BAT545 (pin 2). VDD_P252 is connected to VDD (pin 2) and R668 (pin 1). R668 is connected to COM_P252 (pin 10k). NDB_P252 is connected to NDB (pin 3) and R669 (pin 1). R669 is connected to NDX_P252 (pin 100k). DX_P252 is connected to D633 (pin 1) and NDX_P252 (pin 1). D633 is connected to BAT54 (pin 2). D5L_P252 is connected to D634 (pin 1) and D5L_P252 (pin 1). D634 is connected to US1MH (pin 2). D52_P252 is connected to D635 (pin 1) and D52_P252 (pin 1). D635 is connected to US1MH (pin 2). SW_A2 is connected to D636 (pin 1) and D52_P252 (pin 1). D636 is connected to US1MH (pin 2). The Q616 module is shown with pins 1, 2, 3, and 4. The module is labeled Q616 MMBT3904.215.

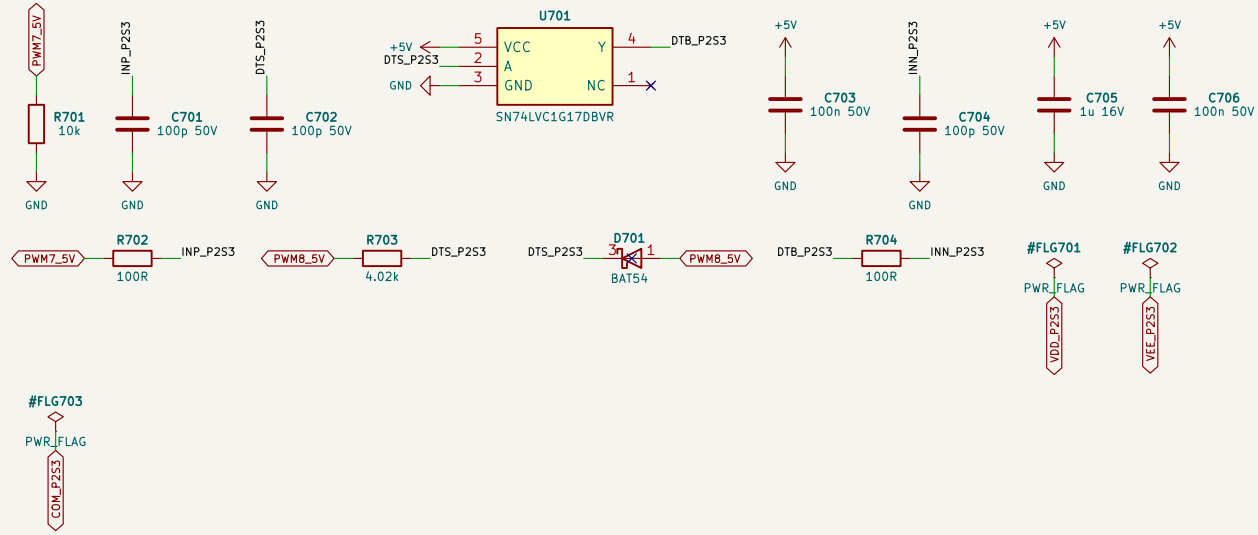
LEG A: LAYOUT RULES

Sheet: /06 Phase 2 leg A gate drive (S1, S2)/

Size: A2	Date: 2026-10-04	Rev: A1
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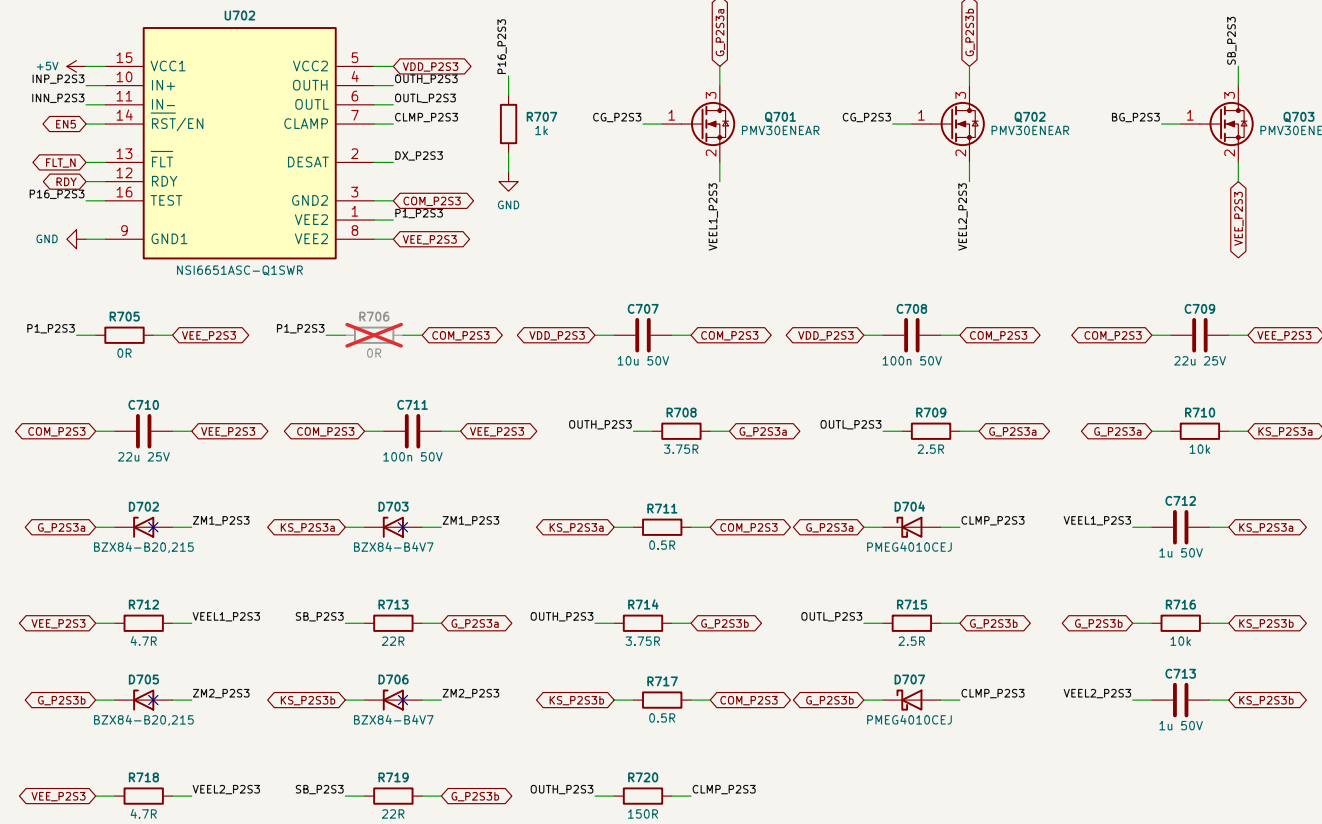
2019	2020	2021	2022	2023
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P2S3: LOGIC SIDE



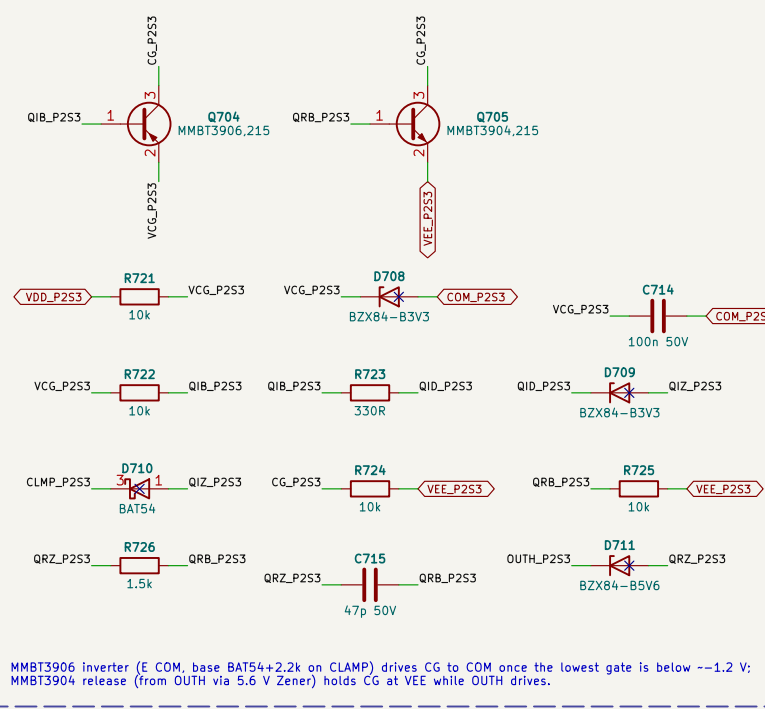
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p
(Interlock + min dead time = max(firmware, hardware)). Bias PG (active low) pulls RDY low.

P2S3: DRIVER AND 2 GATES

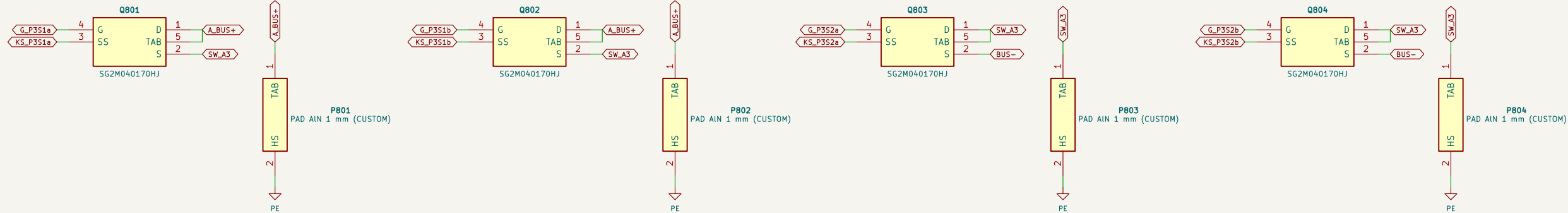


Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.
Miller hold: N-MOSFET clamp (PMV30ENE) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P2S3: MILLER CLAMP CONTROL

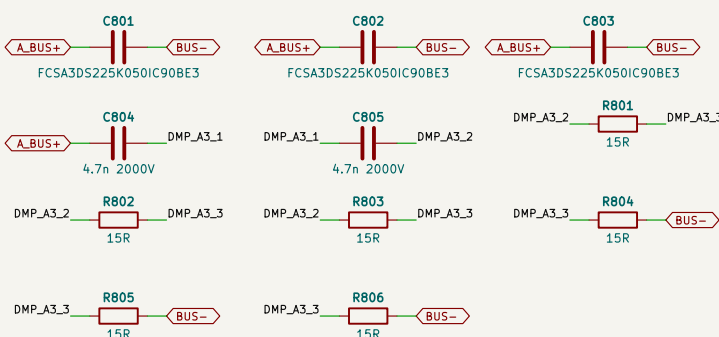


PHASE 3 LEG A: S1 (TOP) / S2 (BOTTOM), 2 X SG2M040170HJ EACH



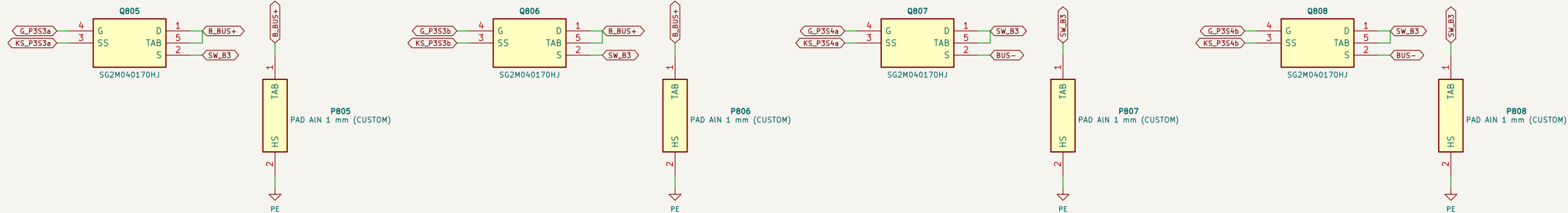
Kelvin source (pin 3) of each device to its own driver COM star (gate-drive sheet). Each tab on its own
AlN ceramic (aluminium nitride) pad to the EARTHED heatsink (basic HV-PE, B2). LAYOUT (IC-22): milled slot between drain pin 1 and
pins 2-4; equal drain / source busbar paths for the two devices of a switch.

PHASE 3 LEG A DECOUPLING + RC DAMPER



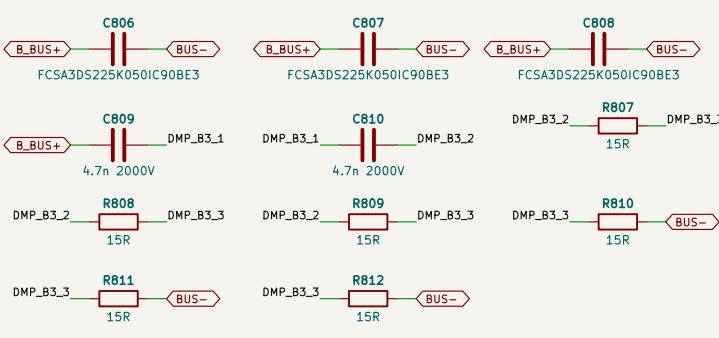
3 x FCSA3DS225 at the S1/S2 pins: commutation loop <= 20 nH, bank <= 15 nH from the leg (LAYOUT, leg deck).
Damper (PVR-02): 2 x (3 x 15R CRCW2512-HP) + 2 x 4.7n 2 kV C0G in series, loop at the pins <= 3 nH.

PHASE 3 LEG B: S3 (TOP) / S4 (BOTTOM), 2 X SG2M040170HJ EACH



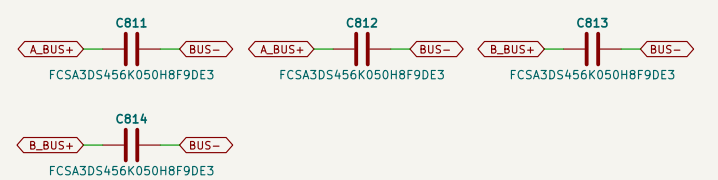
Kelvin source (pin 3) of each device to its own driver COM star (gate-drive sheet). Each tab on its own
AlN ceramic (aluminium nitride) pad to the EARTHED heatsink (basic HV-PE, B2). LAYOUT (IC-22): milled slot between drain pin 1 and
pins 2-4; equal drain / source busbar paths for the two devices of a switch.

PHASE 3 LEG B DECOUPLING + RC DAMPER



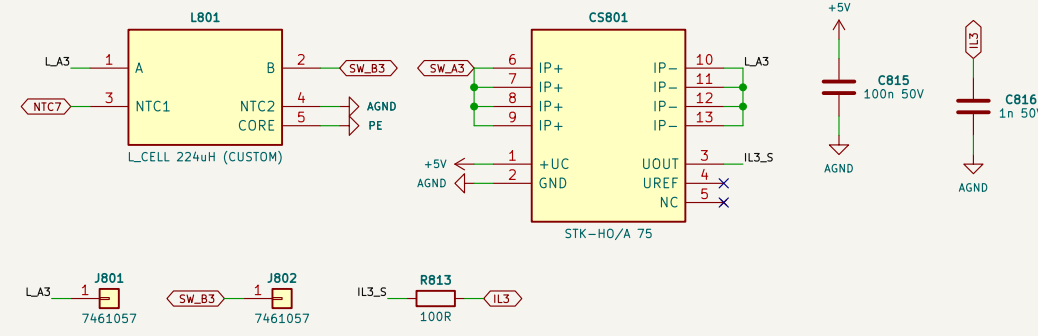
3 x FCSA3DS225 at the S3/S4 pins: commutation loop <= 20 nH, bank <= 15 nH from the leg (LAYOUT, leg deck).
Damper (PVR-02): 2 x (3 x 15R CRCW2512-HP) + 2 x 4.7n 2 kV C0G in series, loop at the pins <= 3 nH.

PHASE 3 SHARE OF THE PORT BANKS



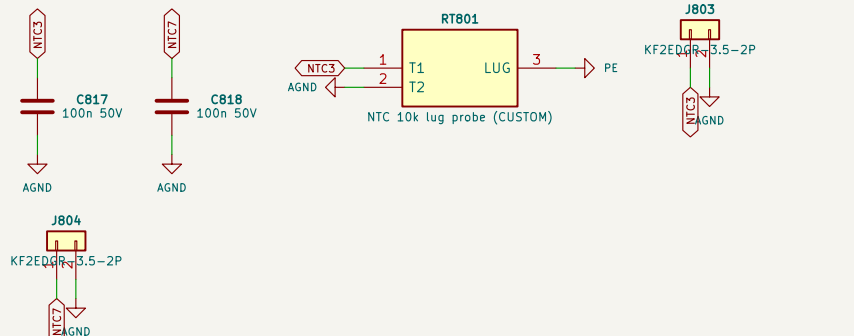
2 x FCSA3DS456 (45 uF) on A.BUS+ and 2 on B.BUS+, at this phase's legs: the banks of all phases form the
port A / port B DC links. Bleeders: one string per port on the port sheets.

PHASE 3 INDUCTOR (CHASSIS, CUSTOM) AND CURRENT SENSOR



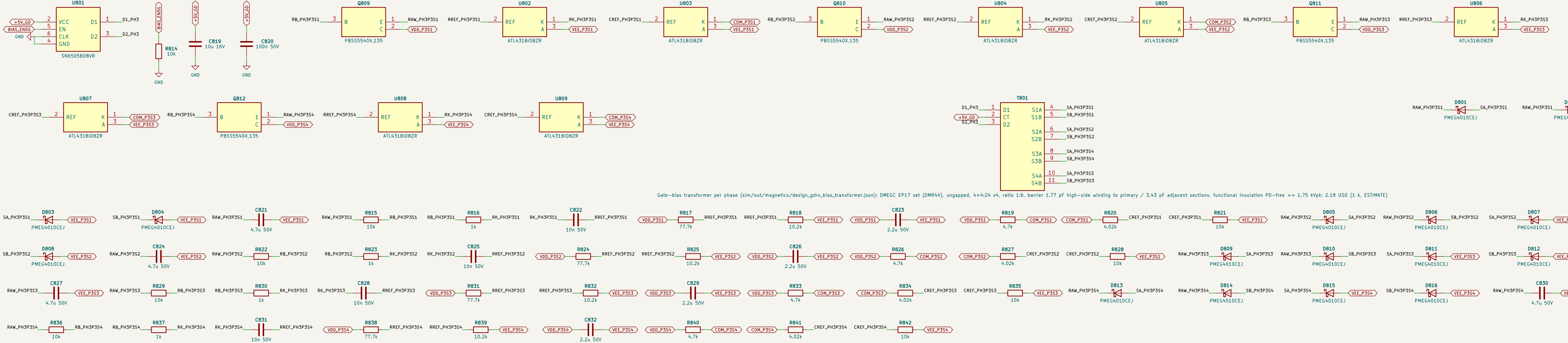
Lead A from SW.A3 passes the STK-HO/A 75 U-primary (pins 6-9 in, 10-13 out: + = SW.A -> SW.B) to stud
LA3; lead B on stud SW.B3. IL3 = 2.5 V + 10.6mV mV/A (Vref 2.48-2.52 V), 0.5-4.5 V over +/-187.5 A;
100R + 1 nF at the source. Primary at the switch node: functional barrier (4 kV rms, 8 kV impulse).

PHASE 3 NTC INPUTS (NTC TO AGND, BIASED ON THE CONTROL BOARD)



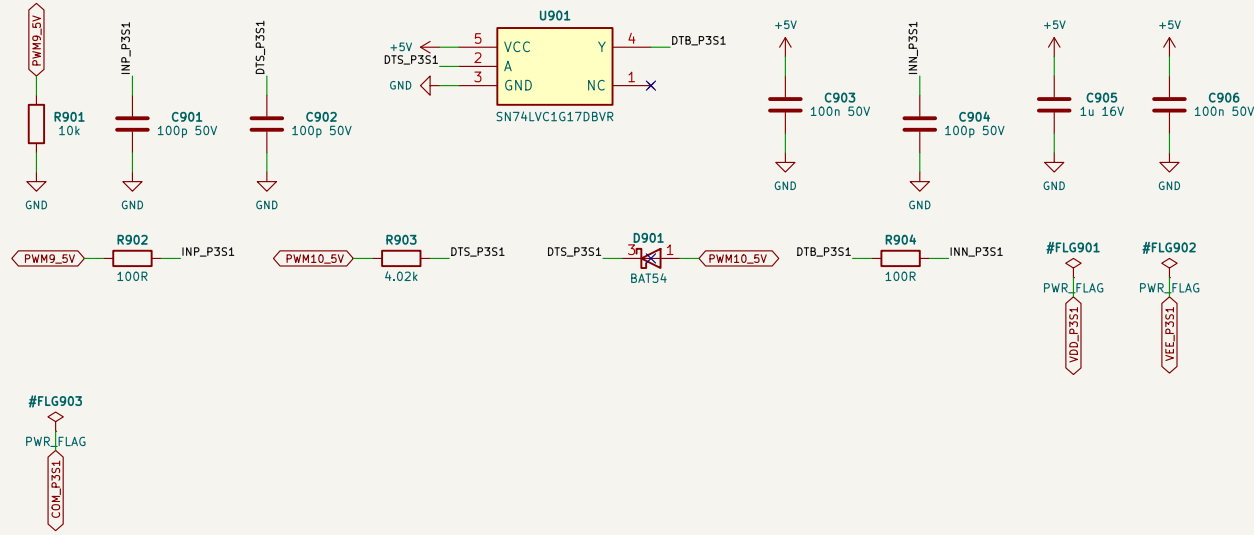
NTC3 = heatsink section of phase 3; lug probe on the EARTHED heatsink, >= 2.5 kV rms lead-to-lug (R-10).
NTC7 = inductor winding NTC (functional to the winding), 100 nF at the header against switch-mode noise.

PH3: GATE BIAS (PER PHASE)



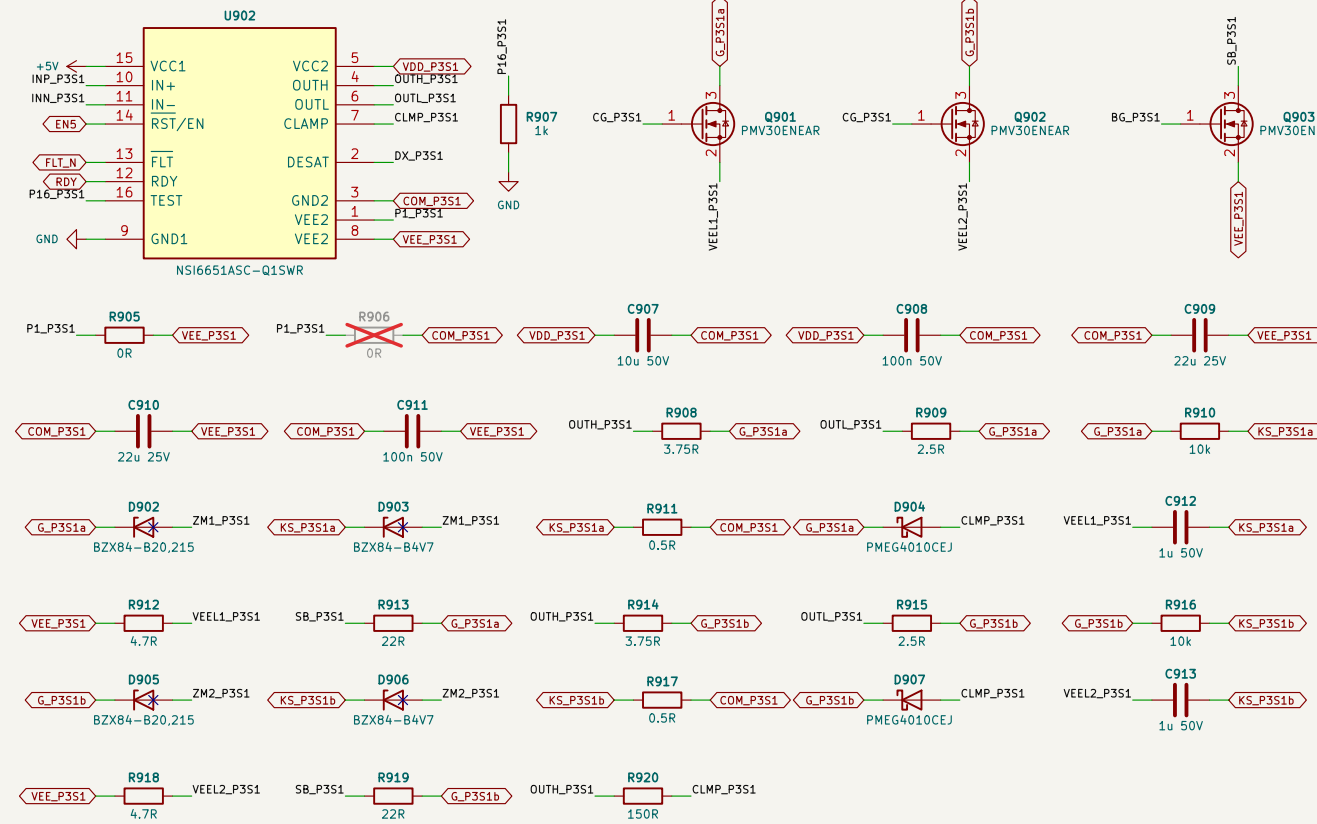
SM6505B push-pull 420 kHz from +5V_GD, BIAS_EN high = on; T_BIAS4 custom, functional insulation
(PD-free >= 1.75 kVpk, <= 5 pF per secondary). Per channel: bridge, PNP regulator VDD-VEE, ATL431 COM split.

P3S1: LOGIC SIDE



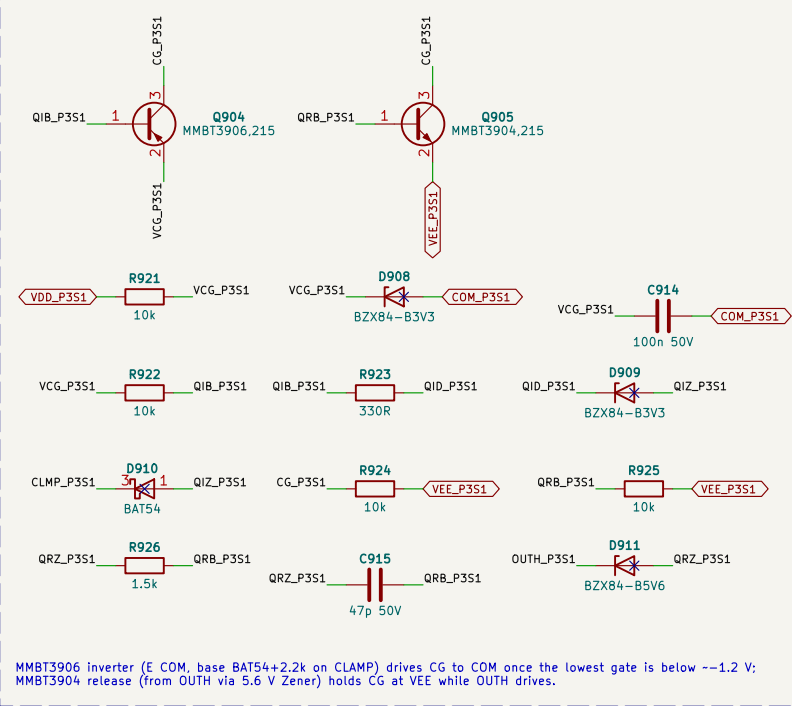
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p
(Interlock + min dead time = max(firmware, hardware)). Bias PG (active low) pulls RDY low.

P3S1: DRIVER AND 2 GATES



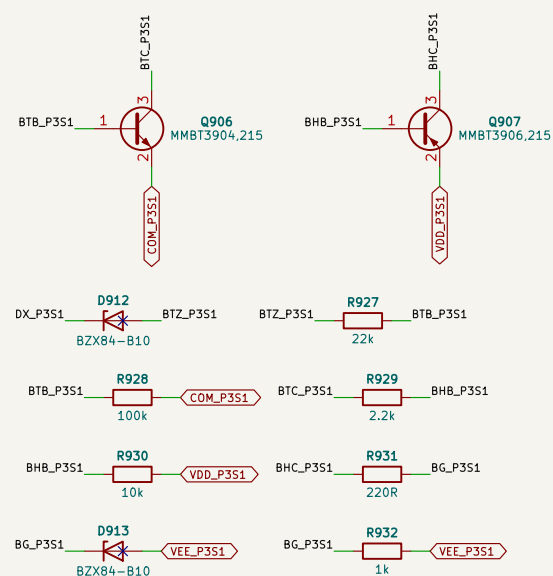
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.
Miller hold: N-MOSFET clamp (PMV30ENE) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P3S1: MILLER CLAMP CONTROL



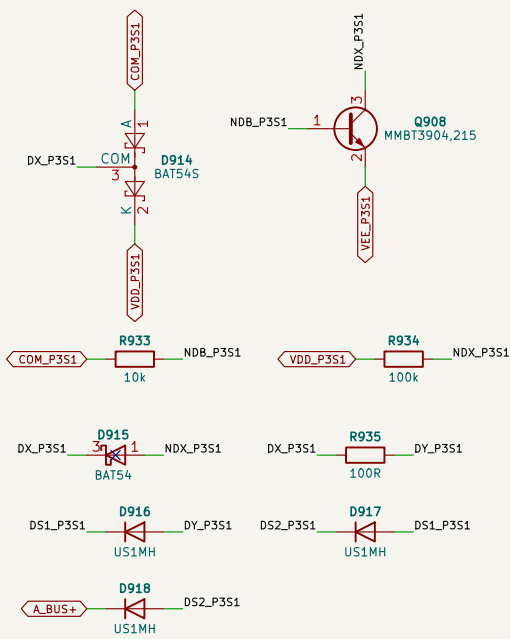
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below --1.2 V; MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P3S1: SHORT-CIRCUIT BOOSTER



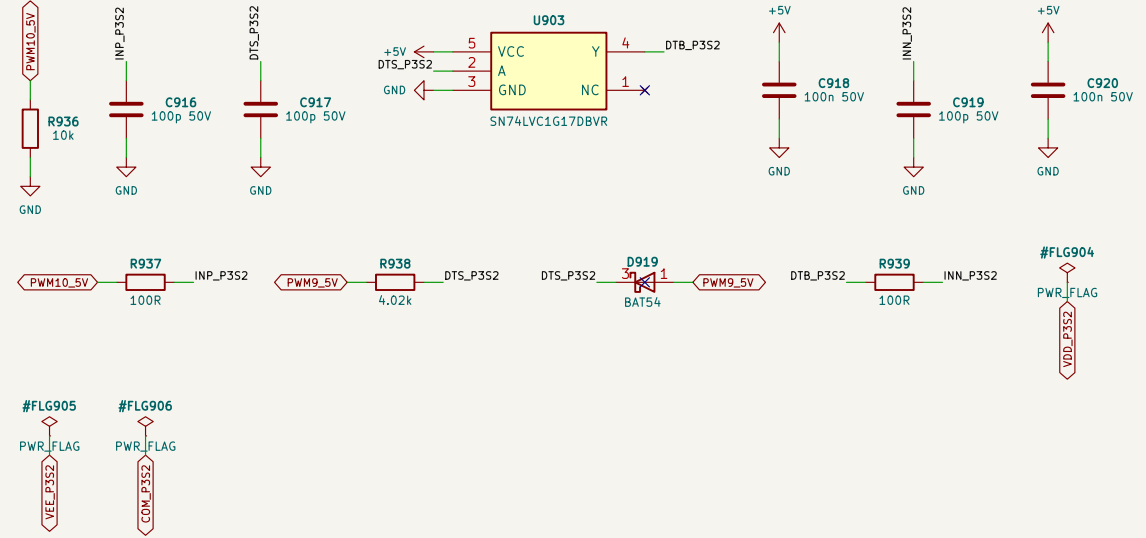
DESAT > 10.3-10.9 V (above the 9.8 V maximum trip) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V; per device AQ3400A + 22R pull the gate to VEE; BG holds ~3 us (1k + FET gates).

P3S1: DESAT



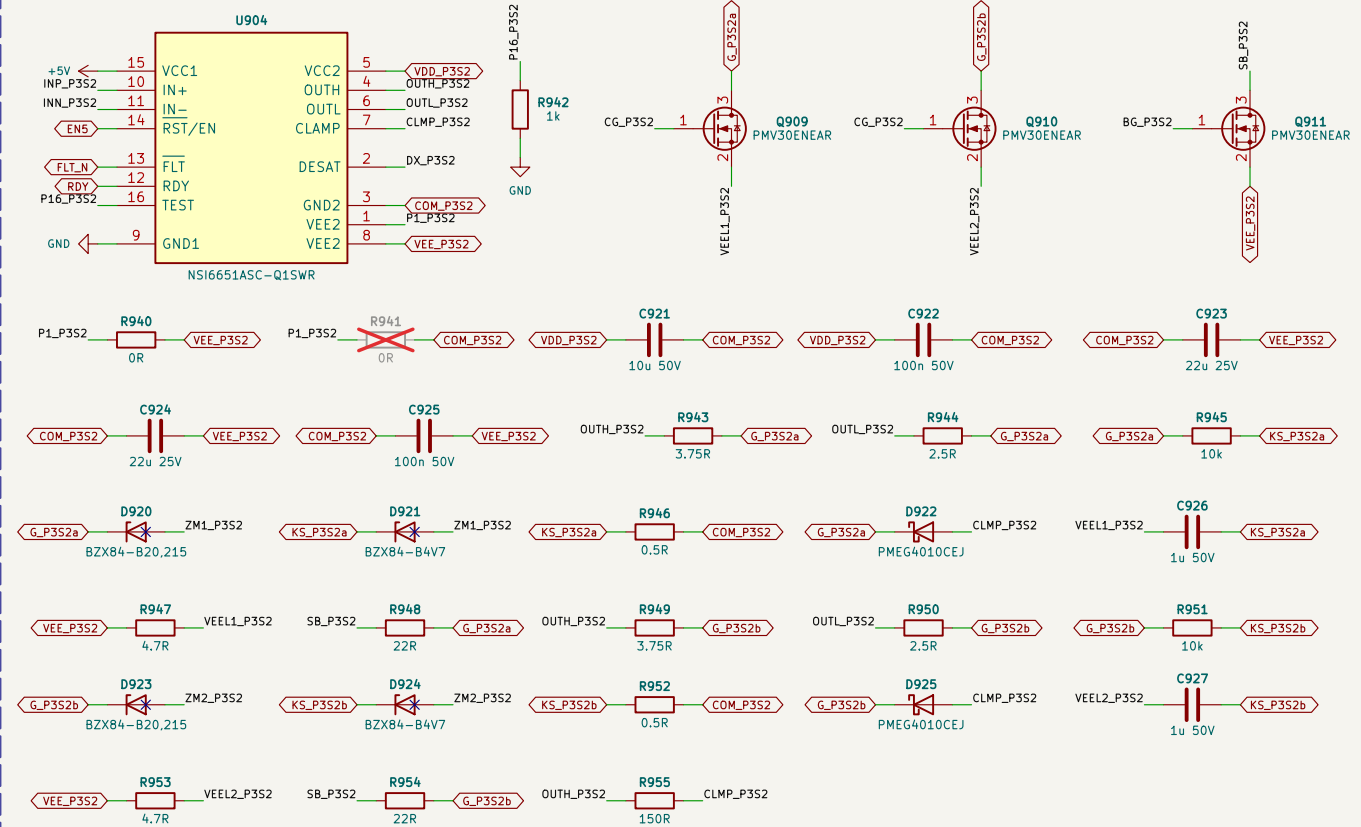
DESAT pin (internal 500 uA blanking source, trip 8.5-9.8 V) - Rs - 3 x US1M - drain, BAT54S keeps DX between COM and VDD against dv/dt through the string; Cdk footprint DNP.

P3S2: LOGIC SIDE



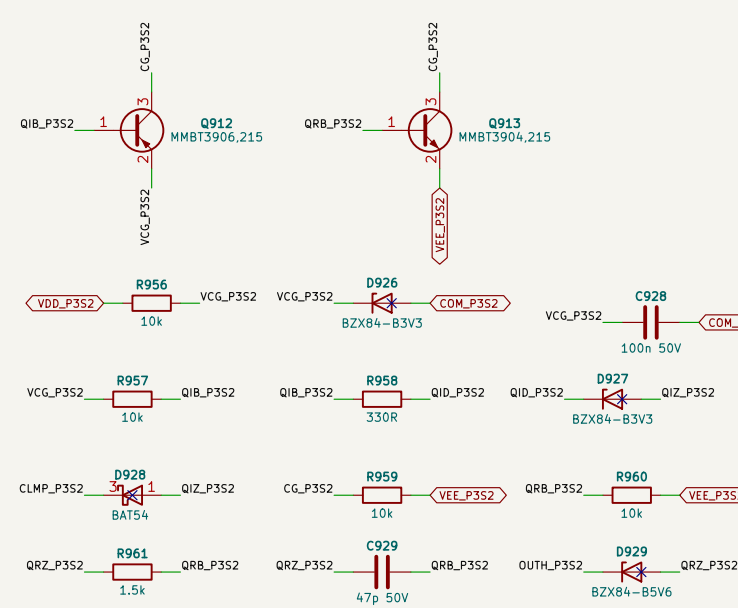
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p
(interlock + min dead time = max(firmware, hardware)). Bias PG (active low) pulls RDY low.

P3S2: DRIVER AND 2 GATES



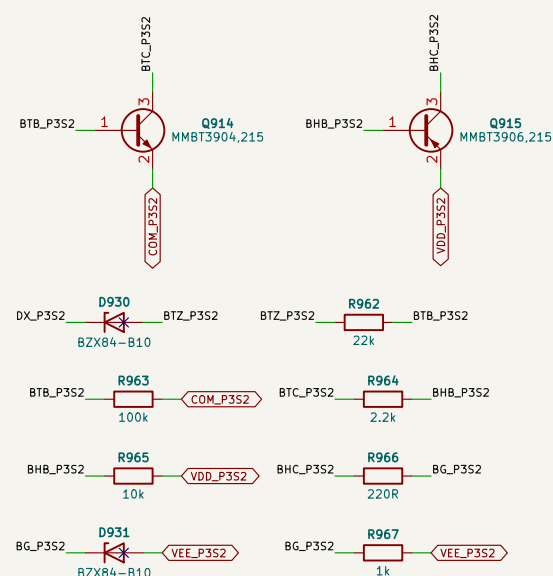
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.
Miller hold: N-MOSFET clamp (PMV30ENE) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P3S2: MILLER CLAMP CONTROL



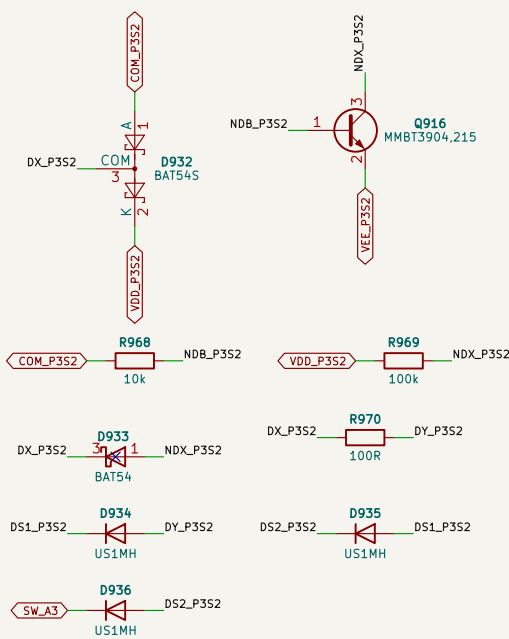
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below --1.2 V; MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P3S2: SHORT-CIRCUIT BOOSTER



DESAT > 10.3-10.9 V (above the 9.8 V maximum trip) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V; per device AQ3400A + 22R pull the gate to VEE; BG holds ~3 us (1k + FET gates).

P3S2: DESAT

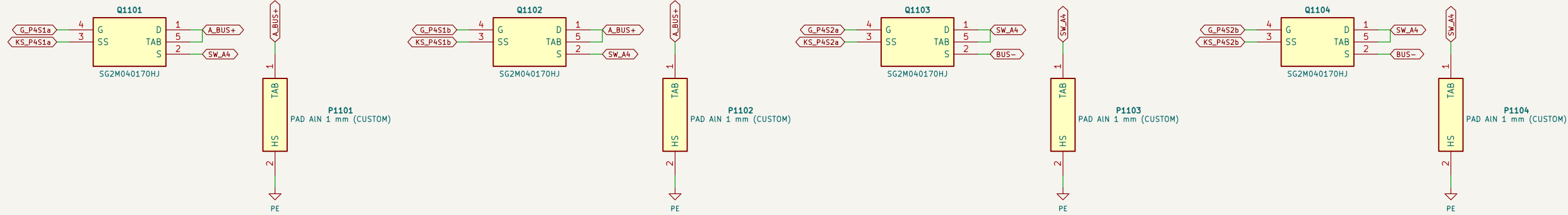


DESAT pin (internal 500 uA blanking source, trip 8.5-9.8 V) - Rs - 3 x US1M - drain, BAT54S keeps DX between COM and VDD against dv/dt through the string; Cdk footprint DNP.

LEG A: LAYOUT RULES

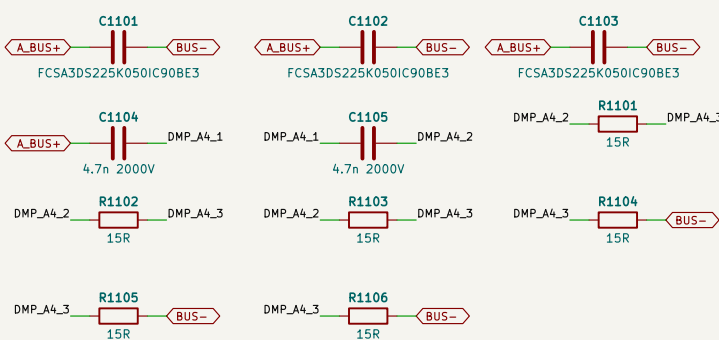
Gate and Kelvin of each device as coupled pairs of equal length (+/-10 %) from the COM star; Ron, Roff, Kelvin R, 10k, Zeners at each device. Clamp FET + C.VL at its device's gate-Kelvin pins, loop <= 1 nH. DESAT string >= 2.5 mm pad gap per diode; board coated to PD1 (IC-22). Miller hold (opposite, zero): die +1.41 V vs VGS(n) min 1.94 V at 170 C - BENCH TEST. Driver input side and bias primary decoupled to GND locally (CM current).

PHASE 4 LEG A: S1 (TOP) / S2 (BOTTOM), 2 X SG2M040170HJ EACH



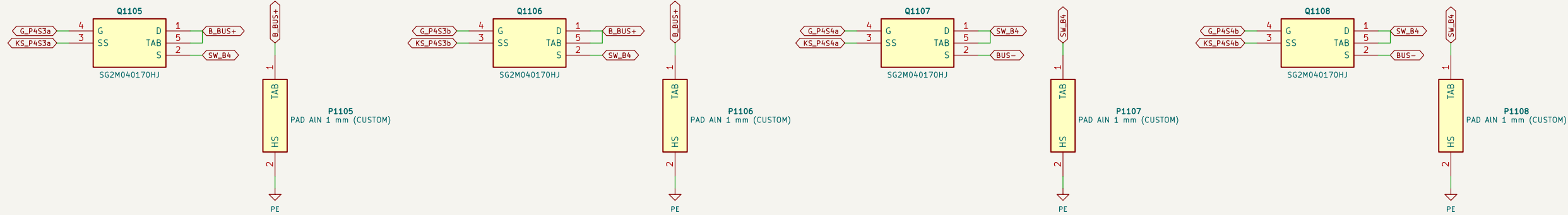
Kevin source (pin 3) of each device to its own driver COM star (gate-drive sheet). Each tab on its own
AIN ceramic (aluminum nitride) pad to the EARTHED heatsink (basic HV-PE, B2). LAYOUT (IC-22): milled slot between drain pin 1 and
pins 2-4; equal drain / source busbar paths for the two devices of a switch.

PHASE 4 LEG A DECOUPLING + RC DAMPER



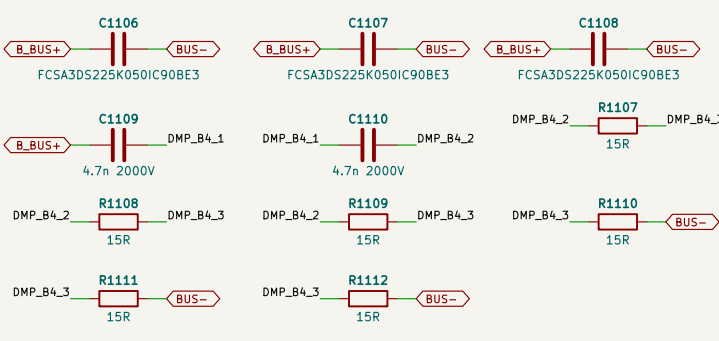
3 x FCSA3D5225 at the S1/S2 pins: commutation loop <= 20 nH; bank <= 15 nH from the leg (LAYOUT, leg deck).
Damper (PVR-02): 2 x (3 x 15R CRCW2512-HP) + 2 x 4.7n 2 kV C0G in series; loop at the pins <= 3 nH.

PHASE 4 LEG B: S3 (TOP) / S4 (BOTTOM), 2 X SG2M040170HJ EACH



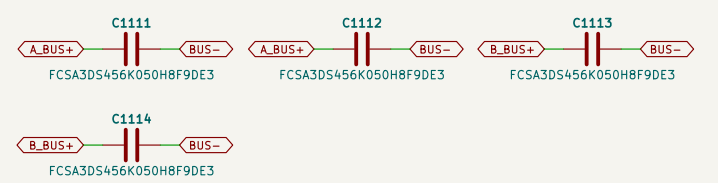
Kevin source (pin 3) of each device to its own driver COM star (gate-drive sheet). Each tab on its own
AIN ceramic (aluminum nitride) pad to the EARTHED heatsink (basic HV-PE, B2). LAYOUT (IC-22): milled slot between drain pin 1 and
pins 2-4; equal drain / source busbar paths for the two devices of a switch.

PHASE 4 LEG B DECOUPLING + RC DAMPER



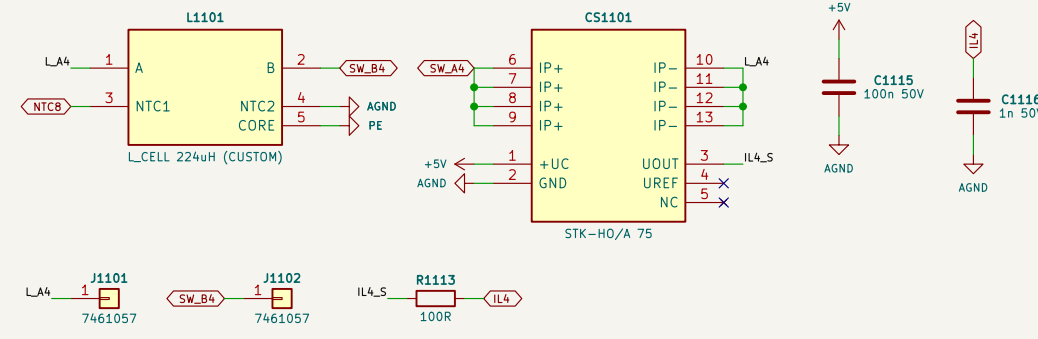
3 x FCSA3D5225 at the S3/S4 pins: commutation loop <= 20 nH; bank <= 15 nH from the leg (LAYOUT, leg deck).
Damper (PVR-02): 2 x (3 x 15R CRCW2512-HP) + 2 x 4.7n 2 kV C0G in series; loop at the pins <= 3 nH.

PHASE 4 SHARE OF THE PORT BANKS



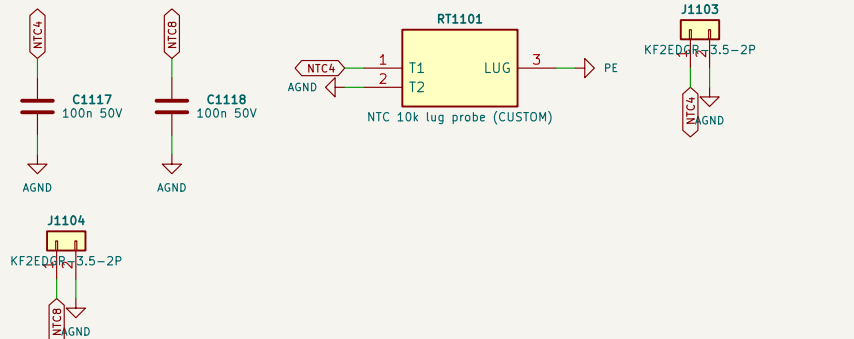
2 x FCSA3D5456 (45 uF) on A_BUS+ and 2 on B_BUS+, at this phase's legs: the banks of all phases form the
port A / port B DC links. Bleeders: one string per port on the port sheets.

PHASE 4 INDUCTOR (CHASSIS, CUSTOM) AND CURRENT SENSOR



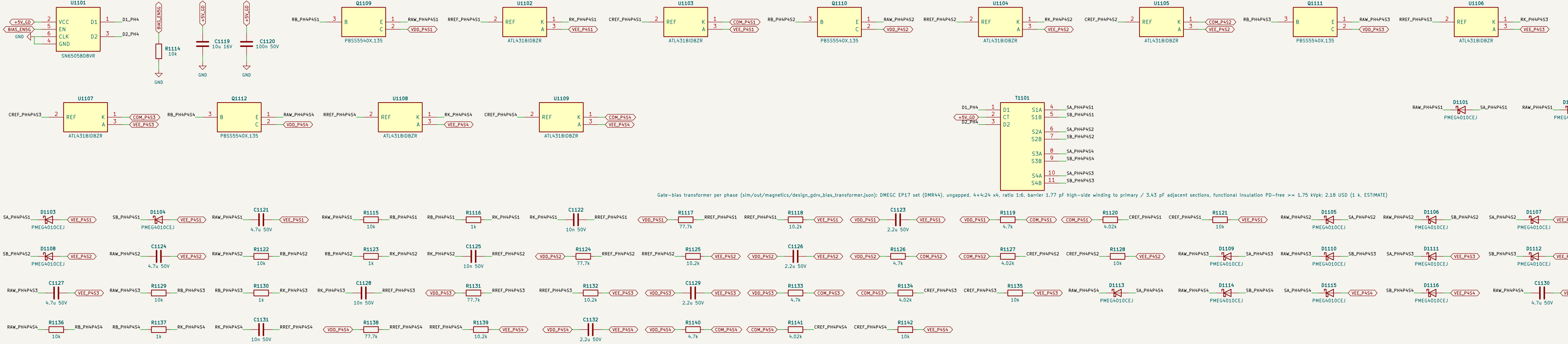
Lead A from SW_A4 passes the STK-HO/A 75 U-primary (pins 6-9 in, 10-13 out; + = SW_A -> SW_B) to stud
L_A4; lead B on stud SW_B4; ILA = 2.5 V + 10.6mV mA (Vref 2.48-2.52 V), 0.5-4.5 V over +/-187.5 A;
100R + 1 nF at the source. Primary at the switch node; functional barrier (4 kV rms, 8 kV impulse).

PHASE 4 NTC INPUTS (NTC TO AGND, BIASED ON THE CONTROL BOARD)



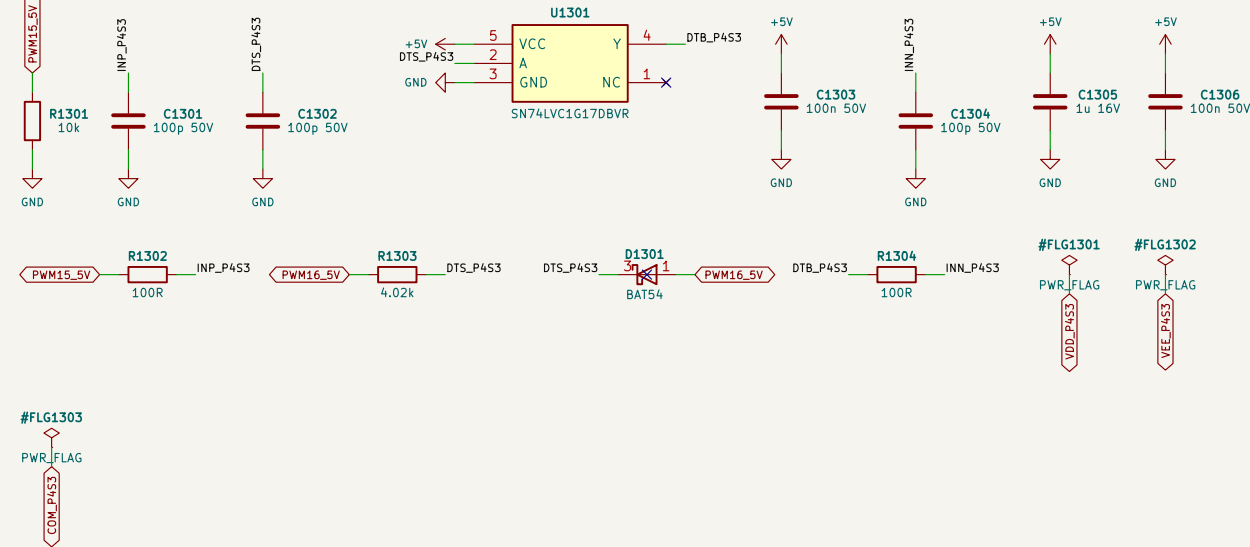
NTC4 = heatsink section of phase 4; lug probe on the EARTHED heatsink, >= 2.5 kV rms lead-to-lug (R-10).
NTC8 = inductor winding NTC (functional to the winding), 100 nF at the header against switch-node noise.

PH4: GATE BIAS (PER PHASE)



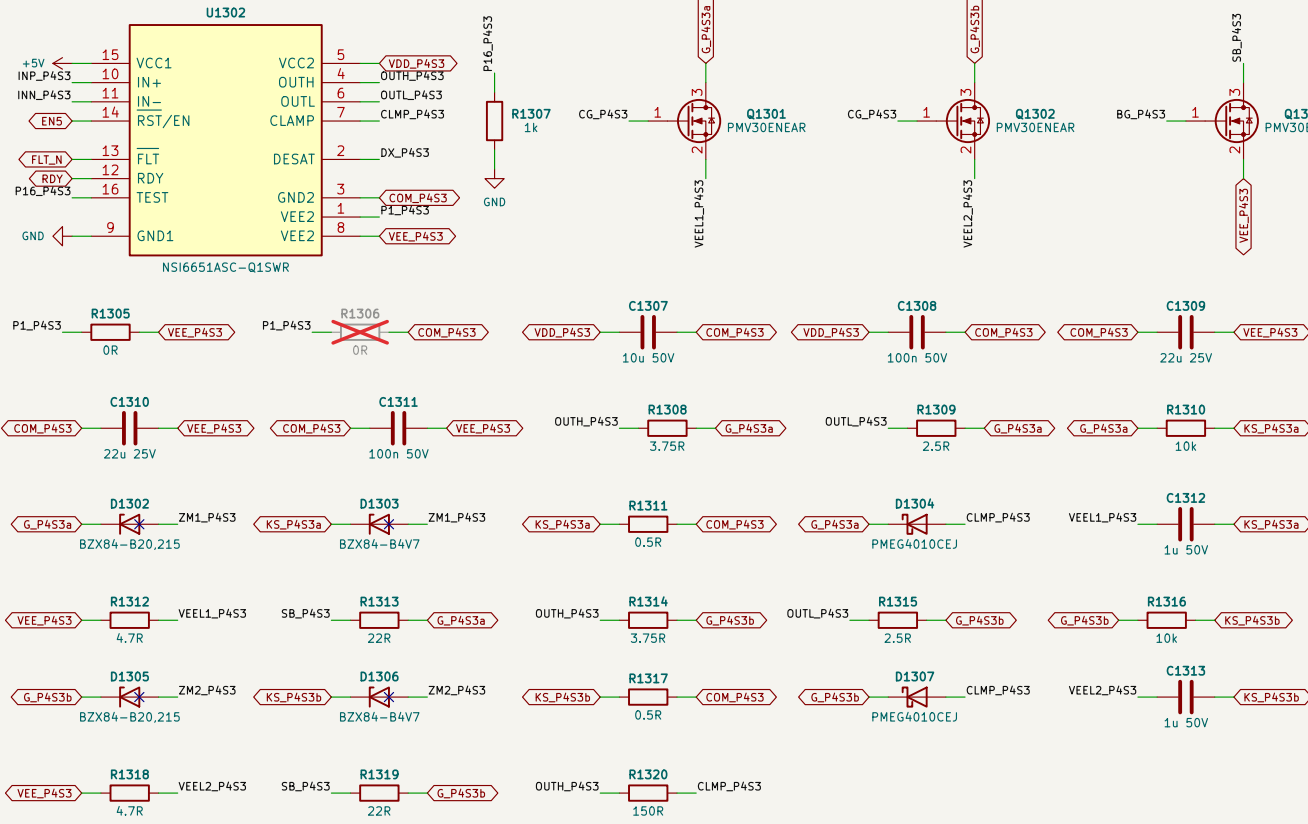
SM6505B push-pull 420 kHz from +5V_GD, BIAS_EN high = on; T_BIAS4 custom, functional insulation
(PD-free >= 1.75 kVpk, <= 5 pF per secondary). Per channel: bridge, PNP regulator VDD-VEE, ATL431 COM split.

P4S3: LOGIC SIDE



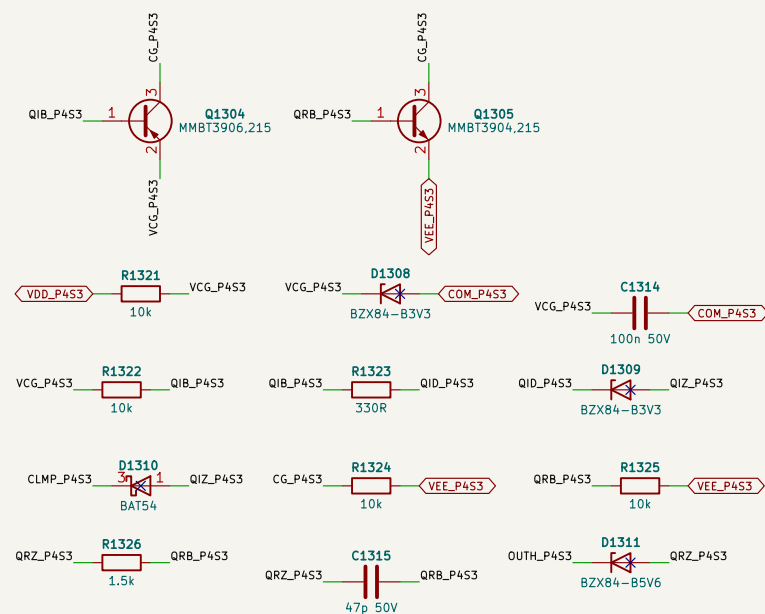
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p (interlock + min dead time = max(firmware, hardware)); Bias PG (active low) pulls RDY low.

P4S3: DRIVER AND 2 GATES



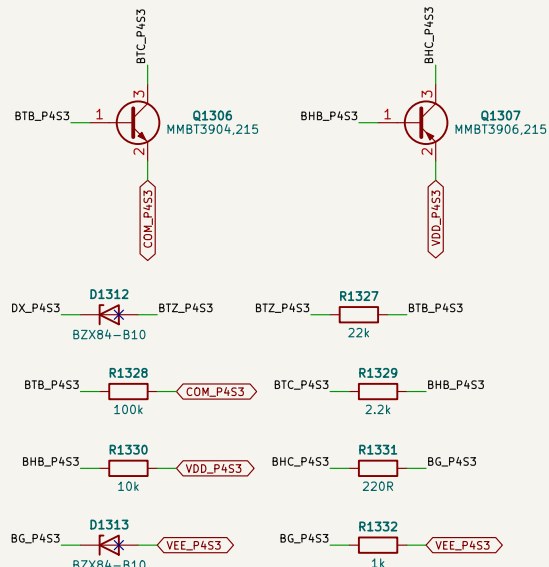
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net. Miller hold: N-MOSFET clamp (PMV30ENEA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P4S3: MILLER CLAMP CONTROL



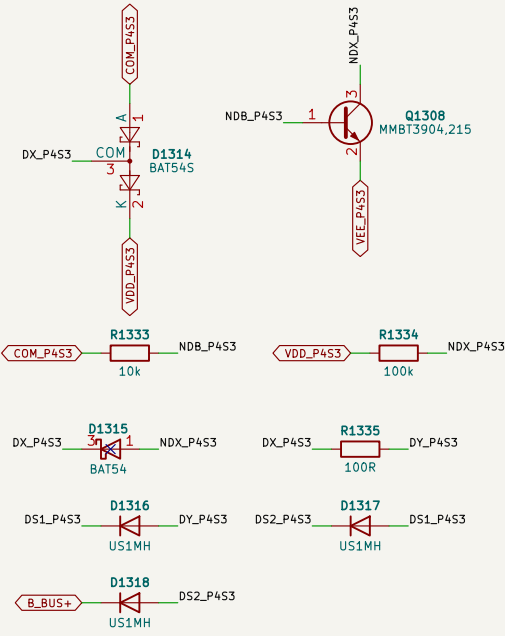
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below -1.2 V; MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P4S3: SHORT-CIRCUIT BOOSTER



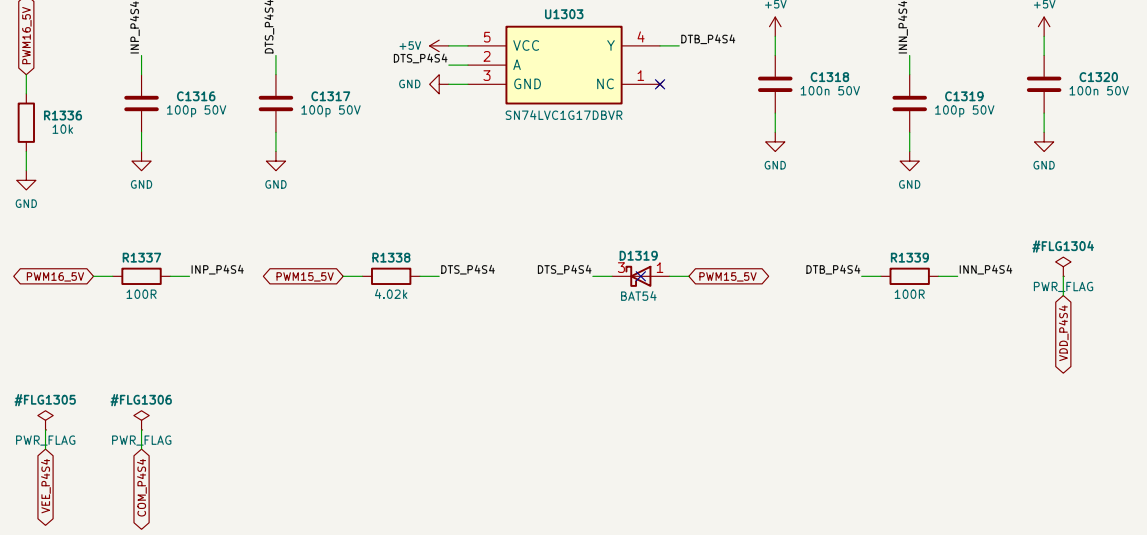
DESAT > 10.3-10.9 V (above the 9.8 V maximum trip) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V per device A03400A + 22R pull the gate to VEE; BG holds -3 us (1k + FET gates).

P4S3: DESAT



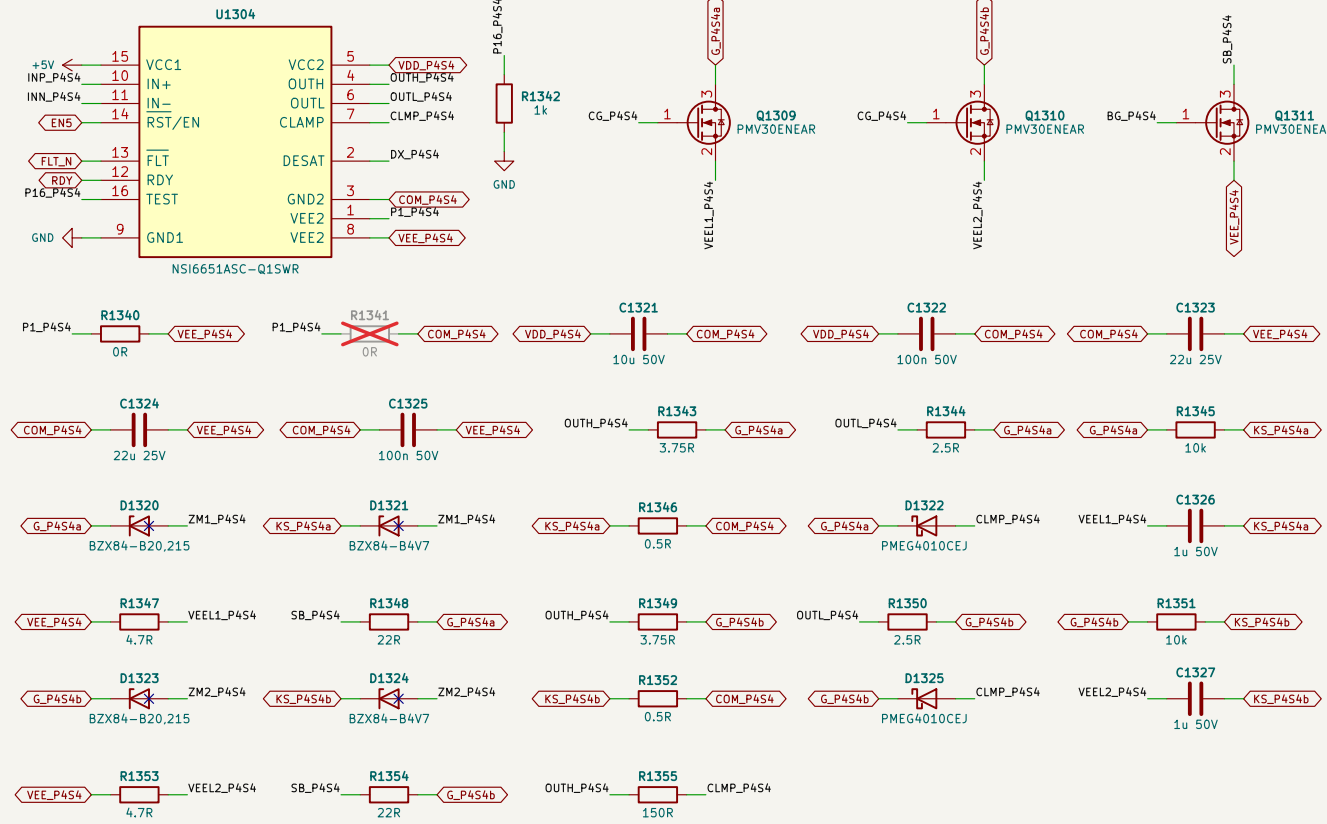
DESAT pin (internal 500 uA blanking source, trip 8.5-9.8 V) - Rs = 3 x US1M - drain, BAT54S keeps DX between COM and VDD against dv/dt through the string; Cdk footprint DNP.

P4S4: LOGIC SIDE



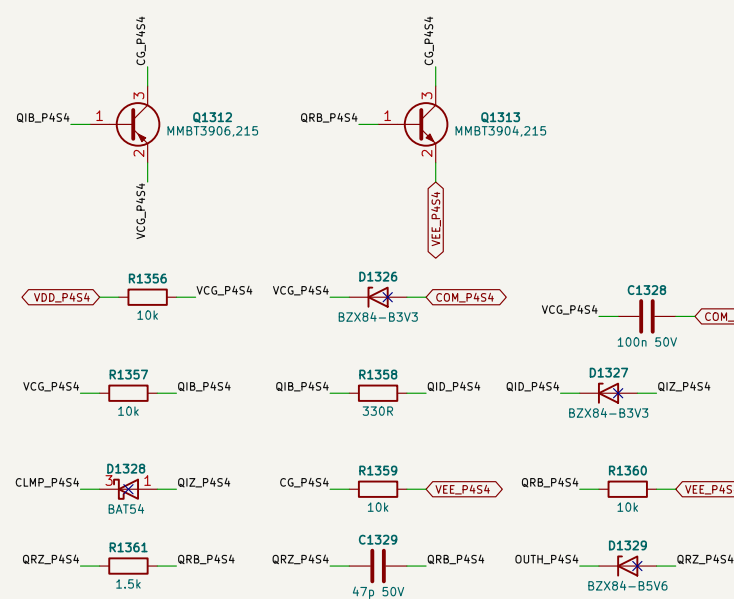
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p (interlock + min dead time = max(firmware, hardware)); Bias PG (active low) pulls RDY low.

P4S4: DRIVER AND 2 GATES



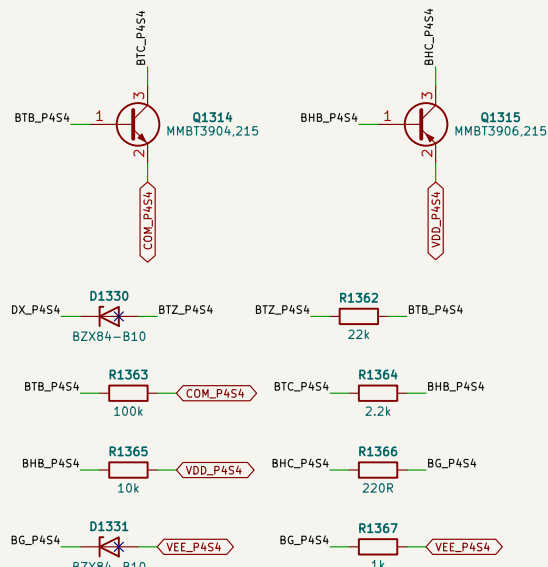
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net. Miller hold: N-MOSFET clamp (PMV30ENEA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P4S4: MILLER CLAMP CONTROL



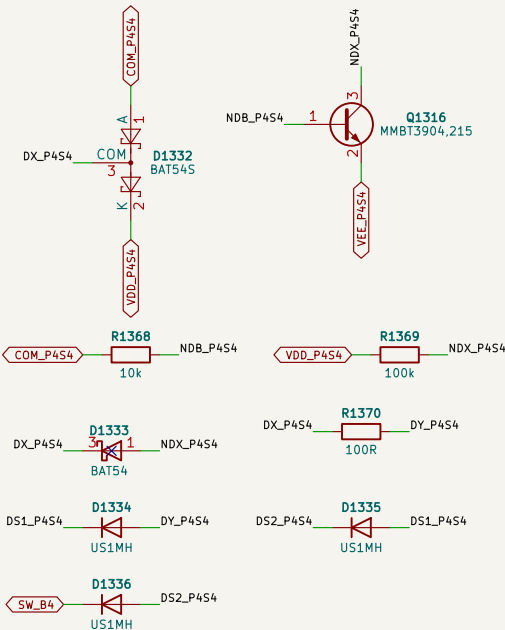
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below -1.2 V; MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P4S4: SHORT-CIRCUIT BOOSTER



DESAT > 10.3-10.9 V (above the 9.8 V maximum trip) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V per device A03400A + 22R pull the gate to VEE; BG holds -3 us (1k + FET gates).

P4S4: DESAT



DESAT pin (internal 500 uA blanking source, trip 8.5-9.8 V) - Rs = 3 x US1M - drain, BAT54S keeps DX between COM and VDD against dv/dt through the string; Cdk footprint DNP.

LEG B: LAYOUT RULES

Gate and Kelvin of each device as coupled pairs of equal length (+/-10 %) from the COM star; Ron, Roff, Kelvin R, 10k, Zeners at each device. Clamp FET + C.VL at its device's gate-Kelvin pins, loop <= 1 nH. DESAT string >= 2.5 mm pad gap per diode; board coated to PDI (IC-22); Miller hold (optical, zero): dle +1.41 V vs VGS(n) min 1.94 V at 170 C - BENCH TEST. Driver input side and bias primary decoupled to GND locally (CM current).

Not bench-validated. Values read from sim/out/*;json; checks: design_check()
Sheet 14
Schematics authored by Chinmoy Bhuyan
Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic

Sheet: /13 Phase 4 leg B gate drive (S3, S4)/
File: 13_ph4-gdb.kicad_sch

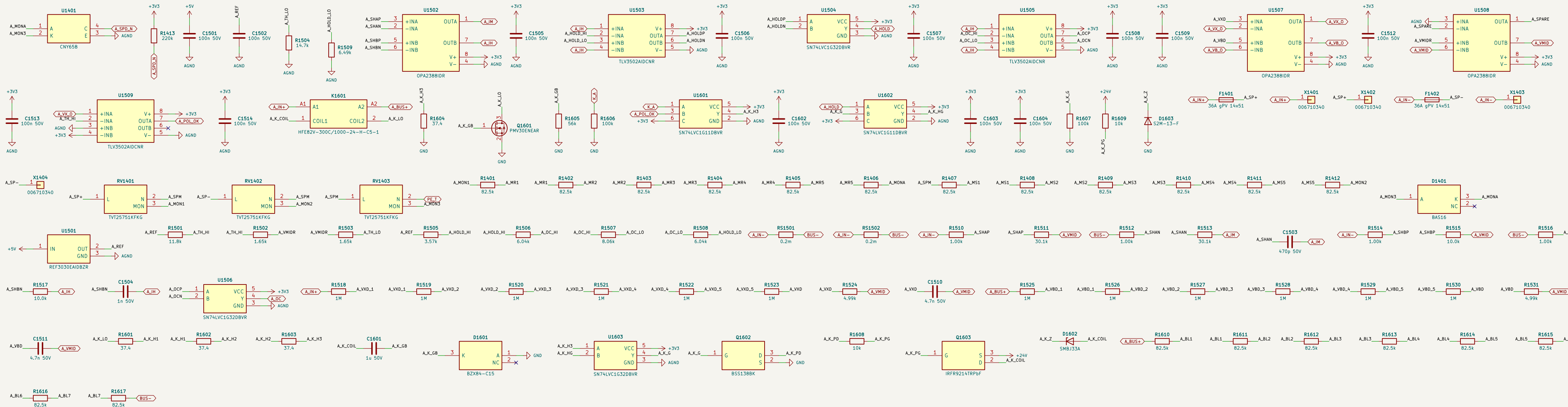
Title: PV-PWR-4 power board - Phase 4 leg B gate drive (S3, S4)

Size: A2 Date: 2026-10-04 Rev: A1
KiCad E.D.A. 10.0.6 Id: 14/26

PORT A (PV SOURCE), D-044 LEAN PORT

Terminal side A_{IN+} / A_{IN-}, bank A_{BUS+}. No fuse, no precharge; array Isc <= 1.25 x rating.
Contactor opens <= 0.97 kA, held above (port_spec lean). Installation: upstream clears 967-1250 A within 0.26 s.

ON-BOARD SURGE PROTECTION: VARISTOR Y (IC-01, IC-12)



TV125751 x 3 (615 VDC, thermal link + monitor lead); MOV1 +/-M, MOV2 +/-M, MOV3 M/PE.T. Up.eff 3793 V pole-PE and pole-pole at In 3 kA (calc. <= 4.0 kV -> reinforced 6 kV / basic 4 kV, D-032); Imax 25 kA. Branch fuses ETI CH14x51 gPV 36 A 30 kA (L/R 2 ms). Monitor loop through both fuses and all three links -> CNY65B -> RB1 (sum on the ID line); RB1 = 1 at rest above 200 V = varistor network open -> CTRL inhibits/stops.

	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F								

Not bench-validated. Values read from sim/out/*.json; checks: design_check() Sheet 16 Schematics authored by Chinmoy Bhuyan Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic		
Sheet: /15 Port A sensing and interlocks/ File: 15_leanA_ctrl.kicad_sch		
Title: PV-PWR-4 power board – Port A sensing and interlocks		
Size: A3	Date: 2026-10-04	Rev: A1
KiCad E.D.A. 10.0.6	Id: 16/26	

	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F								

Not bench-validated. Values read from sim/out/*.json; checks: design_check()
Sheet 17
Schematics authored by Chinmoy Bhuyan
Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic

Sheet: /16 Port A coil drives/
File: 16_leanA_drive.kicad_sch

Title: **PV-PWR-4 power board – Port A coil drives**

Size: A3

Date: 2026-10-04

Rev: **A1**

KiCad E.D.A. 10.0.6

Id: 17/26

Terminal side B_{LN}+ / B_{LN}-, bank B_{BUS}+, HIPE501 250 A wR in both poles (breaks 1.25-50 kA only).
Contactor opens <= 0.97 kA, held above (port_spec lean). Installation: upstream clears 967-1250 A within 0.26 s.

Terminal side B_{LN}+ / B_{LN}-, bank B_{BUS}+, HIPE501 250 A wR in both poles (breaks 1.25-50 kA only).
Contactor opens <= 0.97 kA, held above (port_spec lean). Installation: upstream clears 967-1250 A within 0.26 s.

Figure 1: Schematic representation of the B17L3-220N genome organization and the organization of the B17L3-220N genome organization. The figure shows the genome organization of B17L3-220N and the organization of the B17L3-220N genome organization. The genome is divided into two main regions: the B17L3-220N genome organization and the B17L3-220N genome organization. The B17L3-220N genome organization is shown as a linear map with various genes and their locations. The B17L3-220N genome organization is shown as a circular map with various genes and their locations. The B17L3-220N genome organization is shown as a linear map with various genes and their locations. The B17L3-220N genome organization is shown as a circular map with various genes and their locations.

TVT25751 x 3 (615 VDC, thermal link + monitor lead): MOV1 +/M, MOV2 -/M, MOV3 M/PE/1, Up.eff. 3793 V pole-PE and pole-gate at ln 3 kA (calc., $\leq 4,0 \text{ kV} \rightarrow$ reinforced 6 kV / basic 4 kV, D=032); Imax 25 kA, Branch fuses ETI CH1451 gPV 36 A 30 kA (L/R 2 ms). Monitor loop through both fuses and all three links $\rightarrow$ CNY658 $\rightarrow$ RB1 (sawn on the ID line); RB1 = 1 at rest above 200 V = varistor network open $\rightarrow$ CTRL inhibits/stops.

HPE501/000B100-250 in both poles: breaks 1.25-50 kA only (partial range)
the part OC trip and the contactor keep it out of 250 A-1.25 kA.

HPE501/000B100-250 in both poles: breaks 1.25-50 kA only (partial range)
the part OC trip and the contactor keep it out of 250 A-1.25 kA.

	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F								

Not bench-validated. Values read from sim/out/*.json; checks: design_check() Sheet 19 Schematics authored by Chinmoy Bhuyan Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic		
Sheet: /17 Port B sensing and interlocks/ File: 17_leanB_ctrl.kicad_sch		
Title: PV-PWR-4 power board – Port B sensing and interlocks		
Size: A3	Date: 2026-10-04	Rev: A1
KiCad E.D.A. 10.0.6	Id: 19/26	

	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F								

Not bench-validated. Values read from sim/out/*.json; checks: design_check()
Sheet 20
Schematics authored by Chinmoy Bhuyan
Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic

Sheet: /18 Port B coil drives/
File: 18_leanB_drive.kicad_sch

Title: **PV-PWR-4 power board – Port B coil drives**

Size: A3

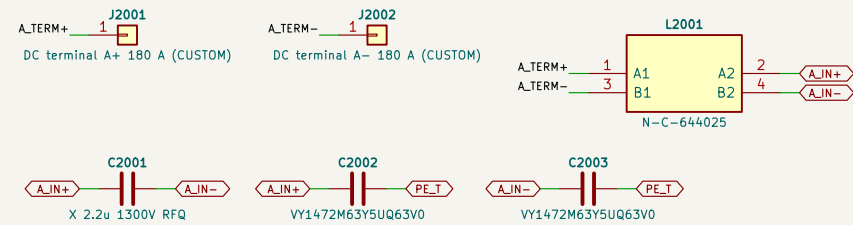
Date: 2026-10-04

Rev: **A1**

KiCad E.D.A. 10.0.6

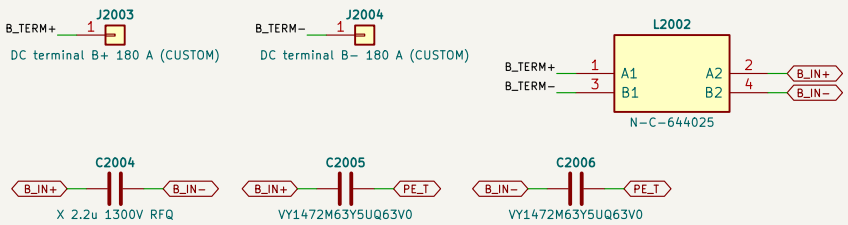
Id: 20/26

PORT A TERMINALS AND EMI FILTER



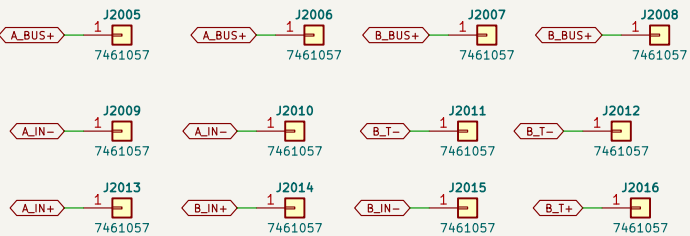
PV: chassis terminals A_TERM+/- (180 A port), both conductors once through the N-C-644025 ring (CM, 150 kHz-30 MHz, ARCHITECTURE 8.1), X 2.2 uF 1300 V at the varistor network, Y1 4.7 nF pole-PE_T (removable PE link for the hipot). Board taps A_IN+ by short links.

PORT B TERMINALS AND EMI FILTER



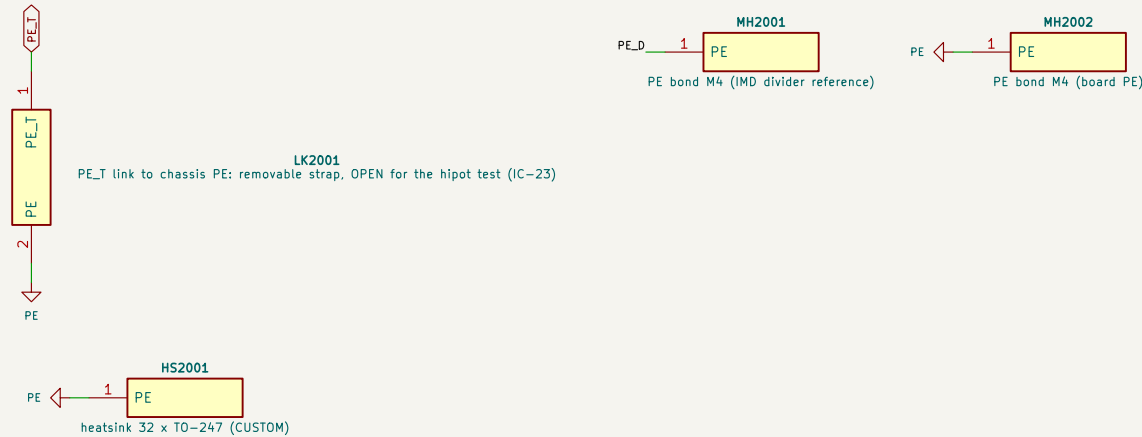
battery: chassis terminals B_TERM+/- (180 A port), both conductors once through the N-C-644025 ring (CM, 150 kHz-30 MHz, ARCHITECTURE 8.1), X 2.2 uF 1300 V at the varistor network, Y1 4.7 nF pole-PE_T (removable PE link for the hipot). Board taps B_IN+ / B_IN- / B_T+ by short links.

BOARD ENTRIES (WURTH REDCUBE PRESS-FIT M3, 100 A EACH)



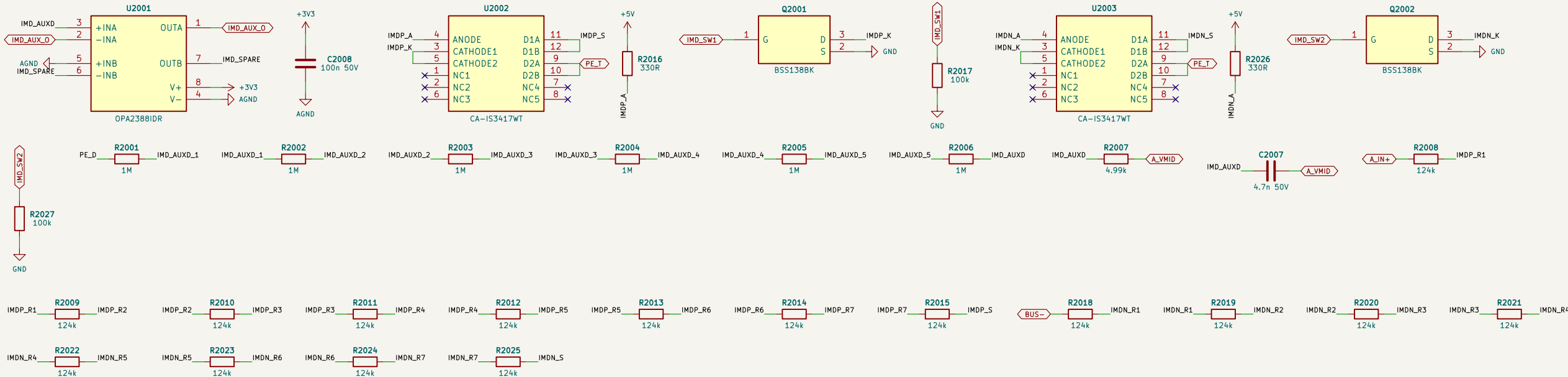
Bus entries from the contactors / fuses: A_BUS+, B_BUS+, A_IN- and B_T- (into the shunts), two studs each; taps A_IN+, B_IN+, B_IN-, B_T+ (varistors, dividers, IMD, precharge, aux input), one each.

PE: LINKS, HEATSINK



PE_T (Y capacitors, MOV3, IMD strings) reaches PE through a removable strap (open for the hipot, IC-23); PE_D = IMD divider bond; the heatsink is earthed at two points.

INSULATION MONITOR (PORT.LEAN_IMD)



IMD_SW1 = string A+ -> PE, IMD_SW2 = string PE -> BUS- (high = connected); VPE = PE divider against BUS- (VMID of port A).

24 V -> +5V (LMR38020, 400 KHZ, 2 A)

U2101
LMR38020FDDAR

24V
+24V
RT_5V
VIN
EN
RT/SYNC
SW
BOOT
FB
PG
GND
EP

8 SW_5V
7 BOOT_5V
5 FB_5V
6 PG_5V
1 GND
9 EP

C2101 4.7u 50V
C2102 4.7u 50V
C2103 100n 100V
R2101 64.9k
L2101 74437368150
C2105 22u 16V
C2106 22u 16V
C2107 22u 16V
R2102 100k
R2103 24.9k
C2104 100n 50V
TP2101
PG_5V

NSI6651 inputs, AHCT, sensors, references, PC +5V, gate bias of phases 1-2, ctrlC2000 circuit (SNVSC40E table 9-1, 400 kHz / 5 V / 2 A): 15 uH, 3 x 22 uF, RT 64.9k, FB 100k / 24.9k 0.1 % -> 4.93-5.10 V. Load in design_check (<= 80 % of 2 A).

gate bias of phases 3–4, ctrlC2000 circuit (SNVSC40E table 9–1, 400 kHz / 5 V / 2 A): 15 uH, 3 x 22 uF, RT 64.9k, FB 100k / 24.9k 0.1 % -> 4.93–5.10 V. Load in design_check (<= 80 % of 2 A).

gate bias of phases 3–4, ctrlC2000 circuit (SNVSC40E table 9–1, 400 kHz / 5 V / 2 A): 15 uH, 3 x 22 uF, RT 64.9k, FB 100k / 24.9k 0.1 % -> 4.93–5.10 V. Load in design_check (<= 80 % of 2 A).

[illegible][illegible]

SELV 24 V TO THE CONTROL BOARD (J2)



AUX-T1 reinforced winding SELV_24V / SELV_0V, cross-regulated 20.3–29.8 V (aux75_spec); full fan speed needs 24.5 V. Full fan speed is only guaranteed while the converter is switching. Fan buck (12–36 V in) + SELV logic on the control board; >= 10 mm from live copper; nothing else of the SELV domain is on this board.

SELV 24 V TO THE CONTROL BOARD (J2)



AUX-T1 reinforced winding SELV_24V / SELV_0V, cross-regulated 20.3–29.8 V (aux75_spec); full fan speed needs 24.5 V. Full fan speed is only guaranteed while the converter is switching. Fan buck (12–36 V in) + SELV logic on the control board; >= 10 mm from live copper; nothing else of the SELV domain is on this board.

Sheet: /19 Live 5 V / 3.3 V, SELV connector/ File: 19_supplies.kicad_sch							
Title: PV-PWR-4 power board – Live 5 V / 3.3 V, SELV connector							
Size: A3		Date: 2026-10-04					Rev: A1
KiCad E.D.A. 10.0.6							Id: 22/26

Sheet 22		
Schematics authored by Chinmoy Bhuyan		
Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE bus		
Sheet: /19 Live 5 V / 3.3 V, SELV connector/ File: 19_supplies.kicad_sch		
Title: PV-PWR-4 power board - Live 5 V / 3.3 V, SELV connector		
Size: A3	Date: 2026-10-04	Rev: A1
KiCad E.D.A. 10.0.6		Id: 22/26

Schematics authored by Chinmoy Bhuyan

Schematics authored by Chinmoy Bhuyan

Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basi

Sheet: /19 Live 5 V / 3.3 V, SELV connector/

File: 19_supplies.kicad_sch

Title: PV-PWR-4 power board – Live 5 V / 3.3 V, SELV connector

Size: A3	Date: 2026-10-04	Rev: A1
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Size: A3	Date: 2026-10-04	Rev: A1
KiCad E.D.A. 10.0.6		Id: 22/26

RICA0 E.D.A. 10.0.0		10: 22/20
	7	8

TAP A_IN+



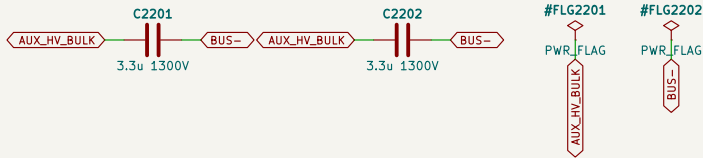
2 per tap AHEAD of the fuse (AC10 pulse-rated, 20 R = the smallest E24 value inside AX-01 / 02: hot-plug I2t 18.1 % of the fuse melting I2t (rule 20 %), film surge 1218 V); 4.14 W each at 200 V / 75 W Fuse: (30 W case, unchanged) per-port fuse 1000 VDC gPV behind 44R: fault current <= 25 A (breaking 20 kA not needed), hot-plug I2t 0.091 A2s = 16 % of melting

TAP B_T+



2 per tap AHEAD of the fuse (AC10 pulse-rated, 20 R = the smallest E24 value inside AX-01 / 02: hot-plug I2t 18.1 % of the fuse melting I2t (rule 20 %), film surge 1218 V); 4.14 W each at 200 V / 75 W Fuse: (30 W case, unchanged) per-port fuse 1000 VDC gPV behind 44R: fault current ≤ 25 A (breaking 20 kA not needed), hot-plug I2t 0.091 A2s = 16 % of melting

SURGE ABSORBER / INPUT FILTER



Film: (30 W case, unchanged) 2 x film = RC surge absorber with 44R: 4 kV / 20 us residual → 1199 V (UR 1300 V); filter corner 546 Hz.

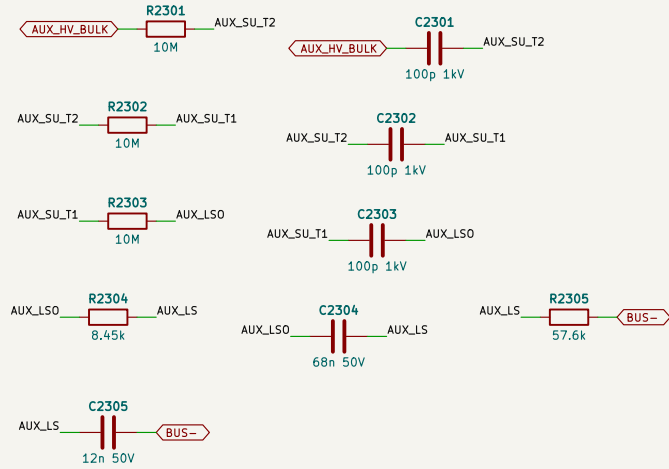
Not bench-validated. Values read from sim/out/*.json; checks: design_check()
Sheet 23
Schematics authored by Chinmoy Bhuyan
Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic

Sheet: /20 AUX75 input taps, OR, film/
File: 20_aux75_input.kicad_sch

Title: PV-PWR-4 power board - AUX75 input taps, OR, film

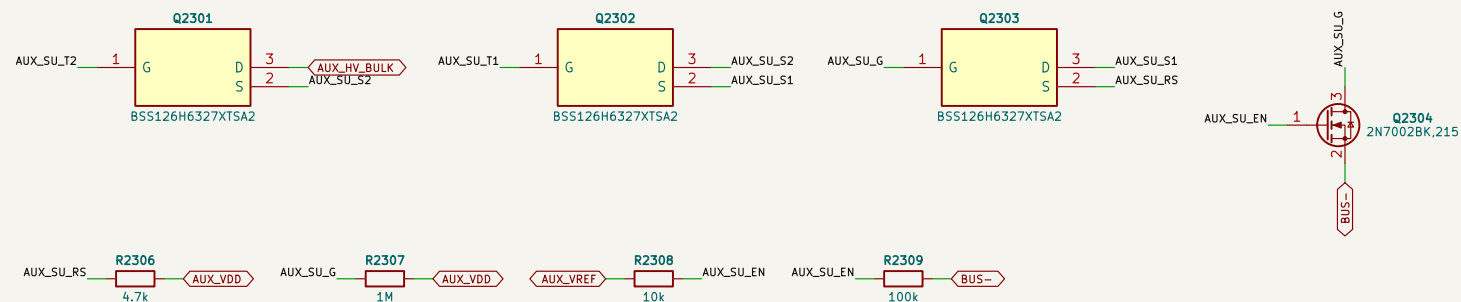
Size: A3	Date: 2026-10-04	Rev: A1
KiCad E.D.A. 10.0.6		Id: 23/26

COMPENSATED TAP STRING



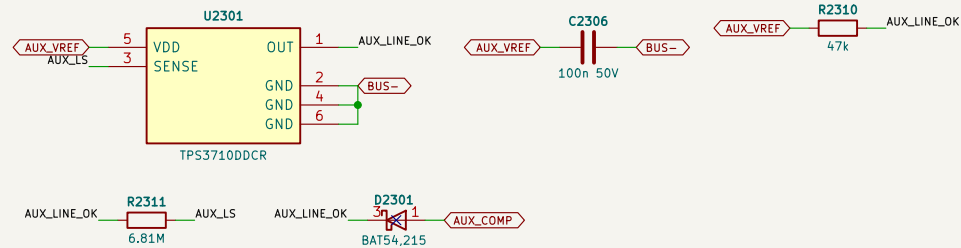
string tap, brown-in 206–215 V, brown-out 179–192 V
brown-in hysteresis (same solve); OV lockout 1107–1162 V

DEPLETION CASCODE CURRENT SOURCE



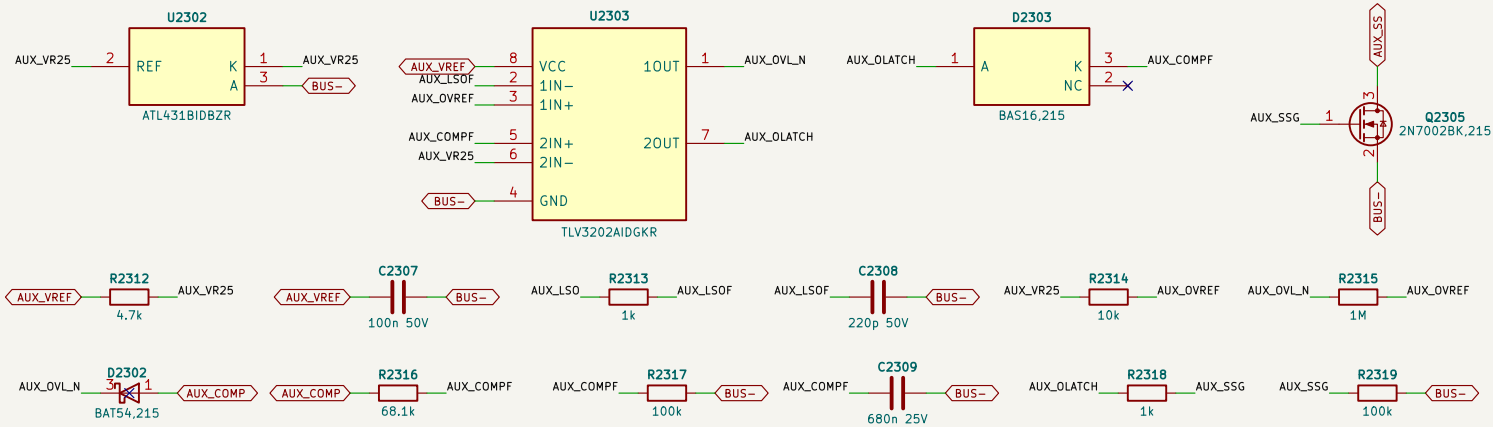
VREF high -> start-up off (as AUX-HV).

LINE BROWN-IN



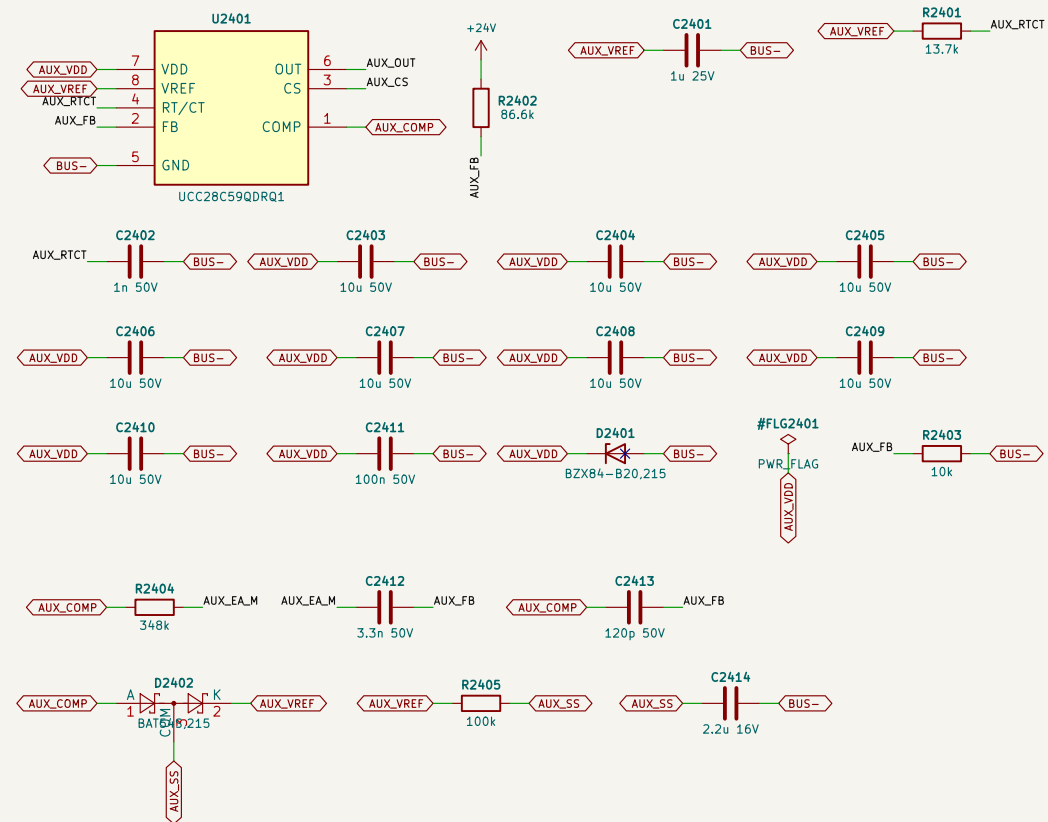
LINE_OK low -> BAT54 holds COMP below the 1.15 V offset.

INPUT OV LOCKOUT AND OVERLOAD TIMER



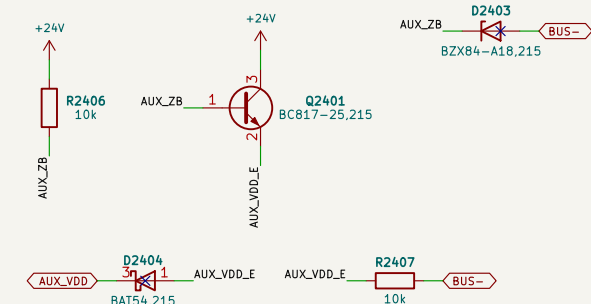
Ch 1: OV lockout -> COMP low. Ch 2: timer divider: trips above 103 / 160 / 256 W (no trip at 75 W + 6 W pull-in); timer 35–80 ms -> latch -> SS crowbar -> hiccup.

PWM CONTROLLER AND ERROR AMPLIFIER



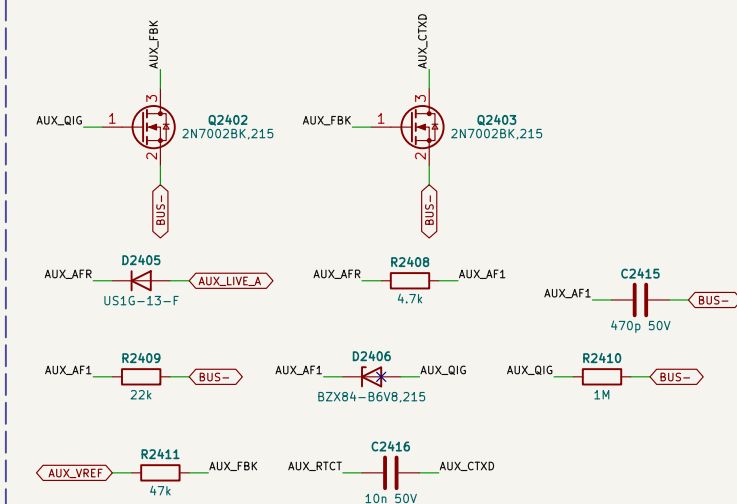
live 24 V into FB: 23.62 / 24.15 / 24.68 V (VFB 2.45–2.55 V, 0.1 %)
type II: mid-band gain 4.0; crossover 241–667 Hz, phase margin ≥ 58 deg type II zero 139 Hz type II pole 3811 Hz

VDD FOLLOWER FROM THE LIVE 24 V



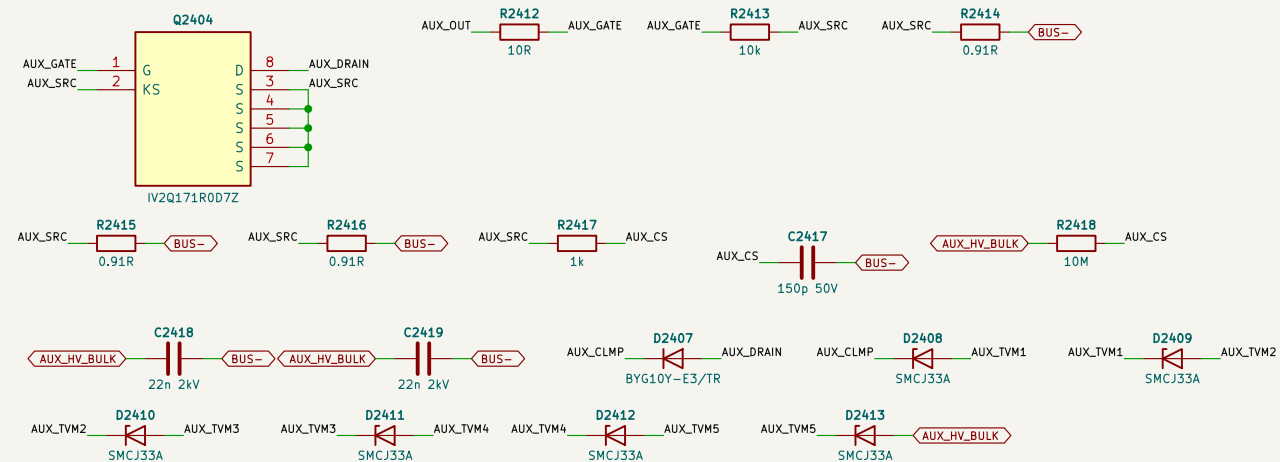
(30 W case, unchanged) follower Zener bias: $V_{DD} = \text{gate drive } 15.46\text{--}18.11 \text{ V}$ (BZX84-A18, $-40\text{--}85 \text{ }^{\circ}\text{C}$)

FREQUENCY FOLDBACK ON THE LIVE WINDING PLATEAU (AX-03)



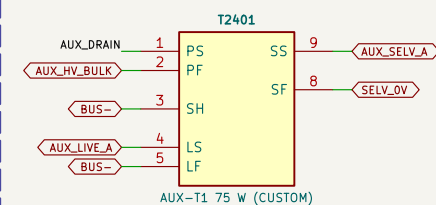
(30 W case, unchanged) foldback: switched onto RT/CT below 7.7–10.1 V output → fsw 5.9 kHz (5.3–6.8)

SIC SWITCH, CURRENT SENSE, LINE FEED-FORWARD, CLAMP



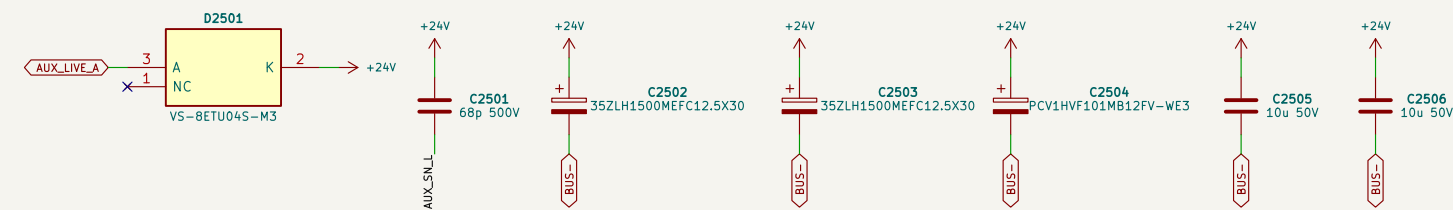
3 In parallel = 0.303333R: lowest limit 2.90 A delivers 103 W ($>= 110\%$ of 75 W) at every input; highest 3.91 A with the line feed-forward (4.27 A without) line feed-forward HV_BULK -> CS (CRH2512): CS offset 101 mV at 1000 V cancels most of the 155-250 nS current delivery overshoot, so the highest current limit no longer rises with the input Clamp: BYG10Y + 6 x SMCJ33A; V_DS 1316 / 1420 V at 1000 / 1100 V.

AUX-T1 (CUSTOM – THE MODULE'S REINFORCED SUPPLY BARRIER B1)



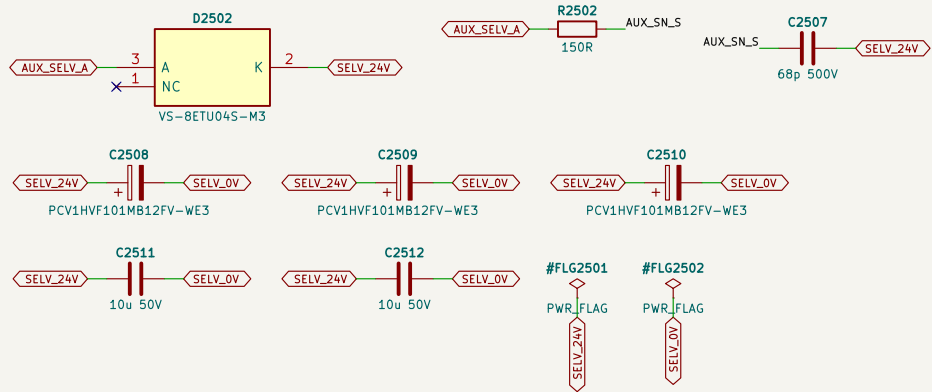
Lp 0.485 mH +/- 7 %; shield and live winding on BUS-; SELV winding reinforced (PD <= 10 pC routine test).
Electrical requirement: aux75_spec.json transformer_requirement.

LIVE 24 V RECTIFIER, HOLD-UP



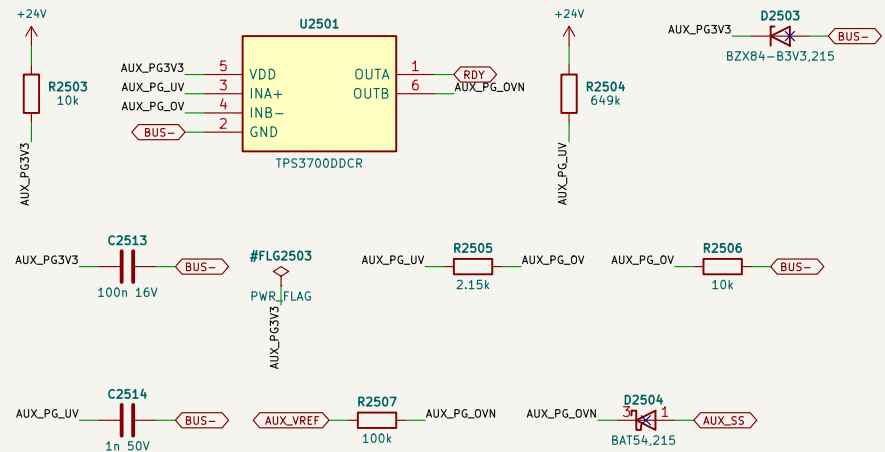
D2PAK tab = cathode on the output. Snubber 150R + 68p.

SELV 24 V RECTIFIER (REINFORCED SIDE)



Cross-regulated from the live winding.

STATUS AND OUTPUT OV STOP (TPS3700 ON A 3.3 V ZENER RAIL)



TPS3700 window on the live 24 V: UV falls 21.02-21.81 V (status), OV rises 26.13-26.76 V (SS pulled low)